How Far Do Arbitrage Frictions Spread?

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Abstract

This paper shows that policies intended to improve a market's efficiency can create frictions that spill back into the same market through shared arbitrageurs. I exploit a novel settlement mismatch created by the 2024 US transition to T+1: US-listed global ETF shares began settling on T+1 while many of their foreign holdings continued to settle on T+2. Relative to non-US ETFs tracking the same indices, these ETFs' primary-market activity becomes less responsive to mispricing, especially in creations. Although US domestic ETFs face no mismatch, their responsiveness declines more when they rely heavily on authorized participants active in affected global ETFs.

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1 Introduction

Arbitrageurs create value[1] by exploiting mispricing between equivalent claims traded in different markets (Gromb and Vayanos 2002, Dávila et al. 2024). These trades connect otherwise segmented markets. But the same trades can also transmit constraints: On September 2, 1998, prices of hurricane bonds, whose payoffs depend on hurricane events, fell by 20% despite no increase in hurricane risk (MacKenzie 2006).[2] That same day, Long-Term Capital Management (LTCM) reported losses on its long Russian government bond–short ruble arbitrage trades following Russia's default. The market suspected the trades were connected through LTCM, which reportedly held hurricane bonds among its other arbitrage positions. Yet another LTCM arbitrage trade, which bet on convergence in the relative stock prices of Royal Dutch and Shell,[3] appears less affected by the announcement.

This contrast raises questions about how arbitrageurs allocate scarce arbitrage capacity across trades: which trades do they preserve, which do they reduce, and which can other arbitrageurs take over? These allocation decisions determine when a constraint on one arbitrage trade impairs another and matter for both investors and policymakers. For investors, trades in an otherwise unaffected market may face weaker price correction because they compete for the same arbitrage capacity. For policymakers, a rule targeting one market may create unintended effects in other markets. Yet these adjustments are difficult to observe because arbitrageurs' positions and trading relationships are largely proprietary.

[Figure 1 here]

I study this question in ETF primary markets, where authorized participants (APs) exploit gaps between ETF share prices and the value of their underlying baskets through arbitrage.[4] ETF primary-market arbitrage (hereafter, "arbitrage") offers a clearly defined and relatively transparent setting for

[1] Arbitrage trades correct mispricing and reallocate risk and payoffs toward investors more willing or able to hold the underlying exposures. For example, suppose one investor values a security at $90 and another at $100. An arbitrageur buying from the first at $94 and selling to the second at $96 earns $2, while each investor gains $4. The trade generates $8 in gains for the investors and $10 in total gains before trading costs.

[2] Hurricane bonds transfer hurricane risk from insurers to investors. Investors receive coupons and repayment of principal unless specified hurricane-related conditions trigger a loss of some or all of their principal.

[3] LTCM was short Royal Dutch Petroleum (Netherlands) and long Shell Transport and Trading (UK). The two stocks represented claims on the same underlying oil business.

[4] This arbitrage is known as creation or redemption: APs create ETF shares by delivering a specified basket of securities and redeem shares for that basket.

studying arbitrage.[5] These markets are also economically large: ETFs reported approximately $6.8 trillion in gross creations and redemptions over fiscal periods ending in 2025, as shown in Figure 1.

Within this setting, I exploit a novel settlement friction triggered by unsynchronized changes in global settlement cycles. In 2024, the US, Canada, and Mexico changed their settlement cycles from two business days after trade (T+2) to one business day after trade (T+1), while other major jurisdictions remained on T+2. This settlement mismatch imposes substantial funding and collateral needs on arbitrage trades spanning two markets, including arbitrage in ETFs tracking foreign assets: US-listed ETF shares began settling on T+1, while many foreign holdings of these ETFs continued to settle on T+2. For domestic ETFs, both their shares and US holdings moved to T+1, so their arbitrage trades face no settlement mismatch. This combination of observed AP links and differential settlement exposure allows me to examine when the friction spreads to domestic ETFs through shared arbitrageurs.

[Figure 2 here]

As Figure 2 illustrates, an AP conducting a redemption must pay for ETF shares on T+1 but receives the proceeds from selling the foreign basket on T+2, creating a temporary funding gap. A creation instead requires the AP to source foreign securities before its T+2 purchases settle, potentially requiring securities borrowing or pre-positioning and associated collateral. Both demands can tie up capacity that the same AP uses to conduct domestic ETF arbitrage. The resources required can be large relative to the arbitrage profit.[6]

I first test whether the settlement mismatch impairs global ETF arbitrage. I use a difference-in-differences (DID) framework that compares US-listed global ETFs with non-US ETFs tracking the same foreign index. The identifying assumption is that, absent the US settlement transition, arbitrage responsiveness to mispricing would have evolved similarly for US-listed and non-US ETFs tracking the same benchmark. Matching funds by benchmark holds their underlying economic exposure fixed, while only the US-listed fund becomes exposed to the new mismatch between the settlement of its shares and that of its underlying assets. I measure arbitrage responsiveness as the response of

[5] First, passive ETFs track specified indices, making the arbitrage pair between ETF shares and the underlying basket well defined. Second, regulatory filings reveal which APs serve each ETF and their creation and redemption activity. These records let me trace connections between arbitrage trades.

[6] For example, exploiting a gross arbitrage return of one basis point may require the AP to finance 100% of the purchase value until the sale proceeds arrive.

flows—the percentage change in ETF shares outstanding—to lagged mispricing.[7] After the transition, the estimated flow–mispricing slope is 0.047 percentage points lower for US-listed global ETFs than for non-US ETFs tracking the same benchmark. At the treated ETFs' average pre-transition size of $3.40 billion, this reduction in the response of shares outstanding has a daily dollar equivalent of approximately $1.6 million for a one-standard-deviation increase in mispricing. The estimated decline becomes larger as the benchmark weight in T+2-settled securities rises.

The effect is also asymmetric: creation-side arbitrage responsiveness declines, while the redemption-side estimate is statistically insignificant. This pattern suggests that the asset-sourcing gap is more pronounced on average. The impairment intensifies during extreme benchmark-flow episodes. On extreme inflow days, the total estimated decline in creation-side arbitrage responsiveness is 0.156 percentage points; on extreme outflow days, the corresponding redemption decline is 0.296 percentage points. At the mean treated-fund size, these estimates correspond to approximately $5.3 million less creation activity and $10.1 million less redemption activity per fund-day, respectively, for a one-standard-deviation increase in mispricing.

I then exploit weekday variation in settlement timing and cross-sectional variation in AP balance-sheet capacity to examine the mechanism. I test whether redemptions are more impaired on Thursdays, when the funding gap extends over the weekend. I also examine whether APs with greater balance-sheet capacity better absorb the asset-sourcing and collateral demands of creations. As expected, redemption-side arbitrage responsiveness declines more when the funding gap extends across the weekend, while greater AP capacity attenuates the creation-side impairment. These findings are consistent with asset-sourcing and funding needs constraining arbitrage. The FX tests provide little evidence for a currency-hedging explanation. Funds respond by tilting their portfolios toward T+1-settled securities, but investor outcomes deteriorate: tracking error rises and secondary-market liquidity declines. Root mean square (RMS) tracking error increases by 5.5 basis points, approximately 11% of its pre-transition mean.

Next, I examine whether the constraint spreads to US-listed domestic ETFs that face no settlement mismatch of their own. Their shares and underlying baskets both settle on T+1, but they often share APs with directly exposed global ETFs. I use a network shift-share design to test whether domestic

[7] ETF shares outstanding change when APs create or redeem shares, while secondary-market trades transfer existing shares between investors. Changes in shares outstanding therefore capture net primary-market activity through which APs exploit mispricing.

ETFs are more affected when they relied more heavily on exposed APs before the transition. Exposed APs are those that traded with US-listed global ETFs before the transition. A one-standard-deviation increase in indirect exposure reduces post-transition arbitrage responsiveness by 0.042 percentage points. Greater AP balance-sheet capacity attenuates the spillover. Netting improves balance-sheet efficiency by reducing the resources required to support trading. Consistent with this channel, better netting attenuates spillovers, especially when APs are more active in global equity ETFs. These findings suggest that shared arbitrageurs transmit constraints selectively, depending on their capacity and competing demands.

I then examine how AP relationships and activity adjust. Within the same global ETF, more-exposed APs are less likely to continue conducting creations or redemptions. Among domestic ETFs where exposed APs lose creation activity, other APs replace a median of approximately 27% of that loss. Stronger potential substitute APs also attenuate spillovers: domestic ETFs whose other authorized APs have greater bank assets or stronger pre-transition activity ranks experience smaller spillovers. Together, these results suggest that other APs partly absorb displaced activity, but do not necessarily restore responsiveness to mispricing.

I use a theoretical framework, building on the segmented-market logic of Gromb and Vayanos (2002) and its application to ETF primary markets by Gorbatikov and Sikorskaya (2021), to organize these patterns. The framework introduces a shared balance-sheet cost across the global and domestic ETF trades of the same AP. A settlement mismatch raises the resources required for cross-border arbitrage and can therefore reduce domestic arbitrage responsiveness. This negative spillover is not automatic: reduced cross-border activity can release capacity, and other APs can replace the affected arbitrageur. The framework predicts weaker propagation when APs have greater balance-sheet capacity and stronger propagation when poor netting coincides with high global exposure. The empirical heterogeneity is consistent with these predictions.

Related Literature. First, this paper shows how differences in settlement timing constrain arbitrage across markets. Existing research emphasizes noise-trader risk (De Long et al. 1990), short-sale constraints (Harrison and Kreps 1978), equity-capital constraints (Shleifer and Vishny 1997), funding constraints (Gromb and Vayanos 2002, Brunnermeier and Pedersen 2008), and intermediary balance-sheet constraints (Pan and Zeng 2016). Related work documents multi-market arbitrage

and persistent price dislocations across integrated markets (Gagnon and Karolyi 2010, Pasquariello 2014). To my knowledge, this is the first paper to show that a settlement mismatch between the two legs of an arbitrage trade reduces arbitrage responsiveness without directly changing underlying asset payoffs. Differences across creations and redemptions, the duration of the settlement funding gap, and AP capacity link this impairment to asset-sourcing and funding needs. These operational demands can tie up capacity that arbitrageurs also use in otherwise distinct trades.

Second, this paper examines how shared arbitrage capacity and substitution across arbitrageurs shape shock propagation through economic networks. Prior work studies transmission through bank–borrower relationships (Khwaja and Mian 2008, Galaasen et al. 2025), production networks (Barrot and Sauvagnat 2016, Carvalho et al. 2021), and product-market competition (Lang and Stulz 1992, Chen et al. ming). Within ETF markets, Gorbatikov and Sikorskaya (2021) and Hong et al. (2022) study mispricing comovement and provide evidence of cross-market spillovers during the COVID-19 episode. I contribute by identifying when a constraint affecting one set of arbitrage trades impairs otherwise distinct trades conducted through shared arbitrageurs. The settlement transition changes the funding and asset-delivery requirements of global ETF arbitrage while leaving domestic ETF arbitrage directly unexposed. Predetermined AP relationships allow me to trace transmission between these trades. Spillovers are smaller when APs have greater balance-sheet capacity and vary with netting relative to trading burden. Other APs partly replace displaced activity, and stronger potential substitutes predict smaller spillovers. These findings show that the reach of a constraint depends on both the capacity of shared arbitrageurs and the ability of other arbitrageurs to absorb displaced activity.

2 Institutional Details and Identification Strategy

In this section, I describe the North American T+1 settlement transition and the DID strategy used to estimate the effects of settlement mismatch.

2.1 The North American T+1 Settlement Cycle Transition

The settlement cycle is the time between trade execution and the final exchange of cash for securities. On May 28, 2024, the United States shortened its settlement cycle from T+2 to T+1. The

transition created a settlement mismatch for arbitrage trades involving US securities settling on T+1 and non-US securities remaining on T+2. In our ETF setting, US-listed ETF shares began settling on T+1, while many foreign securities in their underlying baskets remained on T+2.

[Figure 3 here]

As shown in Figure 3, major foreign markets, including the European Union, Japan, and the United Kingdom, remained on T+2.[8] US-listed global ETF shares therefore began settling on T+1 while many of their foreign holdings continued to settle on T+2. Bridging this mismatch can require additional funding, securities borrowing, or pre-positioning of assets and associated collateral. For a given price–NAV deviation, these additional costs can reduce the profitability of arbitrage and tie up AP capacity used across trades.

The transition provides plausibly exogenous variation in the cost of supplying global ETF arbitrage. The SEC adopted the rule on February 15, 2023, with a compliance date of May 28, 2024. The reform sought to reduce settlement exposure, counterparty risk, and associated margin requirements within the domestic US market.[9] It was not motivated by concerns about global ETF arbitrage or AP balance-sheet capacity. The mismatch arose because the US transition interacted with settlement conventions that remained unchanged in other jurisdictions.

The transition also represents a comparatively narrow operational shock. Systemic crises and broad policy interventions can simultaneously change asset valuations, investor risk appetite, and funding conditions, making the effects of arbitrage frictions and their transmission difficult to isolate. Shortening the settlement cycle changes the timing of cash and securities deliveries without directly changing the underlying securities' contractual payoffs. This provides variation in operational arbitrage costs with less scope for confounding changes in asset fundamentals than a systemic crisis or broad macroeconomic intervention.

2.2 Identification Strategy

To examine the direct effect, I compare US-listed global ETFs exposed to the settlement mismatch with non-US ETFs tracking the same benchmark. Each US ETF and its matched non-US counterpart

[8] Table C1 summarizes settlement cycles in major jurisdictions as of December 2024.
[9] See the DTCC report and SEC final rule for further discussion.

form a fund pair. The baseline DID specification estimates changes in outcome levels:

$$Y_{f,b,t} = \beta_{DID}\, \mathbf{1}(\text{US}_f) \times \mathbf{1}(\text{Post}_t) + \delta\, \mathbf{1}(\text{US}_f) + \alpha_p + \alpha_t + \Gamma \mathbf{X}_{f,b,t-1} + \epsilon_{f,b,t}. \tag{1}$$

where $Y_{f,b,t}$ is the outcome for ETF f tracking benchmark b on day t. The indicator $\mathbf{1}(\text{US}_f)$ equals 1 if the ETF is traded in the US, and $\mathbf{1}(\text{Post}_t)$ equals 1 after the settlement cycle change. The treatment-group indicator allows US and non-US ETFs to differ in average levels before the transition. Matched-pair fixed effects, α_p, absorb persistent pair-level characteristics, including the common benchmark, while day fixed effects, α_t, absorb common daily shocks and the standalone post indicator. The vector $\mathbf{X}_{f,b,t-1}$ contains lagged fund-level controls: log total net assets, log fund age, net-of-fee daily fund returns, the signed daily percentage change in shares outstanding, the Roll (1984) impact ratio as a proxy for ETF illiquidity, and mispricing.[10] All fund-level controls are lagged by one trading day.

The identifying assumption is that, absent the US settlement transition, arbitrage responsiveness would have evolved similarly for the paired funds. Same-benchmark matching holds underlying economic exposure fixed. The reform's narrow operational scope, the lack of statistically significant differential pre-transition changes in responsiveness, the placebo-date estimates, stability across event windows, and the stronger creation-side decline among funds with greater T+2 exposure support this assumption. The FX-capacity estimates do not show the attenuation predicted by the FX-risk explanation, although differences between the capacity groups are imprecisely estimated.

3 Data

In this section, I describe my data and present key stylized facts that motivate the subsequent analyses.

Exchange-Traded Funds This study examines 2024, comparing periods before and after the US settlement-cycle change on May 28. I construct the ETF universe using Morningstar, including both listed and delisted ETFs to reduce survivorship bias. Morningstar also provides benchmark

[10] I include Roll because Bae and Kim (2020) show that ETF-share liquidity affects whether authorized participants correct mispricing. Removing this control leaves the estimate essentially unchanged.

information and daily data on net assets, shares outstanding, returns, prices, and NAVs. I supplement fund characteristics for US-listed ETFs with the CRSP Mutual Fund dataset and obtain US intraday prices from TAQ.[11] To focus on standard passive ETF arbitrage, I restrict the sample to passive ETFs that hold the underlying securities directly. I exclude leveraged, inverse, and actively managed funds, whose arbitrage mechanics may be complicated by leverage, derivative pricing, or management discretion. I also apply geographical and market-size filters. ETFs listed in Canada and Mexico also transitioned to T+1, but I exclude them because these countries have relatively small ETF markets. I exclude ETFs tracking assets in China or India, where settlement conventions vary across securities and differ from the T+2 benchmark. Finally, I retain only ETFs tracking benchmarks with aggregate assets under management (AUM) exceeding $100 million to limit the influence of small benchmarks.

[Table 1 here]

Panel A of Table 1 reports pre-transition characteristics for the full sample of 2,120 ETFs, while Panel B covers 251 ETFs sharing a benchmark across US and non-US listing markets. The direct-effect analysis compares 37 US-listed global ETFs with 67 same-benchmark non-US counterparts. The spillover analysis focuses on US-listed domestic ETFs that face no settlement mismatch, using a common sample of 247 funds with available AP-capacity and network measures.[12]

Authorized Participants Detailed information on authorized participants (APs) is sourced from the N-CEN reporting framework, mandated by the SEC as part of the Investment Company Reporting Modernization initiative.[13] This regulatory requirement ensures that all registered investment funds disclose their counterparties, providing an exhaustive list of APs regardless of whether they engaged in creation or redemption activities during the reporting year. This dataset allows for a granular mapping of the creation and redemption networks that underpin the ETF primary market, facilitating an analysis of how frictions in one segment of the market may propagate to others.[14]

[11] I winsorize mispricing and continuous fund-level controls at the 1st and 99th percentiles within each trading day.

[12] The descriptive samples include ETFs tracking both US and non-US benchmarks. The baseline direct-effect regression contains 34,709 pair-day observations; a fund may appear in multiple matched pairs. The broader spillover specification includes 273 domestic ETFs, while the common sample contains 53,965 fund-day observations. Appendix Table B1 documents sample construction.

[13] On October 13, 2016, the SEC adopted new rules, forms, and amendments to modernize reporting requirements and enhance transparency for registered investment companies. See this SEC report for details.

[14] A growing number of studies leverage this informative reporting to examine ETF intermediation (Gorbatikov and Sikorskaya 2021, Chau et al. 2025).

[Table 2 here]

As shown in Table 2, AP activity is highly concentrated. The table covers 39 AP parent groups with positive activity in the matched US-listed ETF sample during 2018–2024. Bank of America, JPMorgan Chase, and Goldman Sachs together account for approximately $12.29 trillion in cumulative creations and redemptions, or 53.0% of recorded activity. Groups classified as banks account for 83.0% of aggregate activity, alongside non-bank APs such as Citadel, Jane Street, and Virtu. APs differ in how they allocate activity across domestic and global ETFs. Their shared relationships provide a potential transmission channel: settlement-related demands from global ETF arbitrage can tie up capacity used to serve domestic ETFs. Section 4.5 examines when these connections transmit the constraint.

To capture AP balance-sheet capacity, I manually identify the parent company of each AP and obtain regulatory data from Capital IQ. I primarily use the Tier 1 capital ratio as a proxy for balance-sheet capacity. This ratio equals Tier 1 capital relative to risk-weighted assets (RWA), so a higher value indicates stronger regulatory capitalization and, as an empirical proxy, greater capacity to absorb the capital and liquidity demands associated with the settlement mismatch.[15]

4 Empirical Results

I first examine the direct effects of settlement mismatch on US-listed global ETFs. I then study when the constraint spreads to domestic ETFs through shared APs and how other APs absorb the resulting adjustments.

4.1 The Direct Result: More Dispersed Mispricing

I first compare the mispricing of US-listed global ETFs with non-US ETFs tracking the same index. Because their markets close at different times, I align the US ETF's price with its non-US counterpart's market close. The available data do not report the US ETF's intraday NAV at that time, so I use the counterpart's local-close NAV as a same-benchmark valuation anchor to calculate synchronized mispricing (hereafter, "mispricing") for the US ETF.[16]

[15] Tier 1 capital includes Common Equity Tier 1 capital and qualifying Additional Tier 1 instruments. Other regulatory measures, such as the total capital ratio, liquidity coverage ratio, and total loss-absorbing capacity ratio, are also used.

[16] Appendix D details the construction.

[Figure 4 here]

Figure 4 compares mispricing distributions before and after the transition, subtracting each group-period mean. The left panel shows that mispricing among US-listed global ETFs becomes more dispersed, and Pearson kurtosis rises from 3.32 to 9.60 ($p < 0.001$). The right panel shows that the change in kurtosis for non-US ETFs tracking the same benchmarks is not statistically significant. These descriptive patterns motivate the following tests of whether primary-market activity becomes less responsive to mispricing.

4.2 The Direct Result: Lower Arbitrage Responsiveness

4.2.1 Baseline Arbitrage Responsiveness

I estimate the following triple-difference regression:

$$\begin{aligned} \text{Flow}_{f,b,t} = {} & \beta_1 \mathbf{1}(\text{US}_f) \times \mathbf{1}(\text{Post}_t) \times \text{Mispricing}_{f,t-1} + \beta_2 \mathbf{1}(\text{US}_f) \times \mathbf{1}(\text{Post}_t) \\ & + \beta_3 \mathbf{1}(\text{Post}_t) \times \text{Mispricing}_{f,t-1} + \beta_4 \mathbf{1}(\text{US}_f) \times \text{Mispricing}_{f,t-1} \\ & + \beta_5 \text{Mispricing}_{f,t-1} + \beta_6 \mathbf{1}(\text{US}_f) + \Gamma \mathbf{X}_{f,b,t-1} + \alpha_p + \alpha_t + \epsilon_{f,b,t}. \end{aligned} \tag{2}$$

The dependent variable, $\text{Flow}_{f,b,t}$, is the signed percentage change in shares outstanding for ETF f tracking benchmark b on day t. Positive values denote net creations, and negative values denote net redemptions. I lag mispricing by one trading day so that the deviation precedes day-t flow. The coefficient β_1 measures the differential post-transition change in arbitrage responsiveness for US-listed global ETFs relative to their matched non-US counterparts.[17] The control vector $\mathbf{X}_{f,b,t-1}$ includes lagged fund size, fund age, returns, the dependent variable, and the Roll (1984) illiquidity measure. Matched-pair fixed effects, α_p, absorb persistent pair-level characteristics, while the US-fund indicator allows average treated–control level differences. Day fixed effects, α_t, absorb shocks common to all funds on a given day, including market-wide changes in liquidity and volatility. The unit of observation is the fund-by-counterpart pair-day; Section 3 describes the sample construction.

[Table 3 here]

[17] The coefficient measures reduced-form arbitrage responsiveness to observed mispricing. Identifying a structural arbitrage-supply elasticity would require additional assumptions about investor demand and the determination of mispricing.

Column (1) of Table 3 shows that estimated arbitrage responsiveness declines after the settlement-cycle change. The triple-interaction coefficient is −0.047 (p = 0.004). A one-standard-deviation increase in mispricing is therefore associated with a 0.047-percentage-point smaller change in shares outstanding for US-listed global ETFs relative to non-US ETFs tracking the same benchmark.[18] For an average fund size of $3.4 billion, this decline represents approximately $1.6 million less daily net flow and about 9% of the standard deviation of daily percentage changes in shares outstanding (0.53%).[19]

4.2.2 Asymmetry between Creation and Redemption

Creation and redemption expose APs to different settlement constraints. When an ETF trades at a premium, creation-side arbitrage requires the AP to source and deliver the basket of non-US underlying securities. After the transition, the US-listed ETF leg settles on T+1, while the underlying securities continue to settle on T+2. This asset gap requires the AP to borrow or pre-position the ETF shares or underlying securities and to commit collateral in the interim.[20] When an ETF trades at a discount, redemption-side arbitrage creates a funding gap: the cash outflow for the US-listed ETF shares settles before the cash inflow from selling the non-US underlying securities. APs can finance this gap through repo, money markets, or dealer credit, but the cost increases when the gap lasts longer or funding capacity is scarce.

To separate the two directions of arbitrage, I construct a maximum-premium measure and a minimum-discount measure over the five trading days preceding day t:

$$\text{Mispricing}^{+}_{f,t} = \max_{\tau \in [t-5,\, t-1]} \{\max(\text{Mispricing}_{f,\tau}, 0)\},$$

$$\text{Mispricing}^{-}_{f,t} = \min_{\tau \in [t-5,\, t-1]} \{\min(\text{Mispricing}_{f,\tau}, 0)\}.$$

[18] The finding is robust to alternative constructions and weighting schemes. Applying the conventional end-of-day price–NAV measure symmetrically to treated and control funds produces a coefficient of −0.037 (p = 0.053), compared with the baseline estimate of −0.047. Estimating the scaling factor from pre-transition observations produces a more negative and statistically significant coefficient. Collapsing the panel to one treated-fund observation per day produces an estimate of −0.068 (p = 0.001). Restricting the sample to European matches, for which trading hours overlap, produces an estimate of −0.054 (p = 0.006).

[19] The dollar amount expresses the estimated change in net-flow responsiveness.

[20] An ETF trust may, by mutual agreement, allow an AP to deliver the underlying securities after the standard deadline, but the trust typically requires collateral against the outstanding obligation. For example, the American Century ETF Trust and the DBX ETF Trust require collateral of 115%.

The window allows for the possibility that APs respond with a delay because sourcing a foreign basket takes time. Ending the window at $t-1$ keeps both measures predetermined.[21]

Columns (2) and (3) of Table 3 report separate estimates for creation-side and redemption-side arbitrage. For premiums, the triple-interaction coefficient is −0.083 ($p < 0.001$), indicating lower creation-side arbitrage responsiveness after the settlement change. For discounts, the coefficient is −0.006 ($p = 0.783$). The decline is concentrated in creation-side arbitrage, suggesting that the settlement mismatch raises asset-sourcing and collateral needs. This interpretation is consistent with industry evidence that APs would need to post additional collateral to complete creations before foreign basket securities settle.[22] The baseline redemption estimate shows no statistically significant average decline in arbitrage responsiveness. However, the following tests show that redemption-side arbitrage responsiveness declines when financing the settlement gap is more costly.

The creation estimate is economically meaningful. At the mean fund size, the premium coefficient of −0.083 has a daily dollar equivalent of approximately $2.8 million for a one-standard-deviation increase in the premium measure and equals about 16% of the standard deviation of daily percentage changes in shares outstanding. The creation-side estimates generally become more negative as settlement-mismatch exposure rises.[23] Figure 5 reestimates β_1 on increasingly restrictive subsamples defined by the share of each ETF's benchmark that remains on T+2. In Panel A, the estimated reduction in creation-side arbitrage responsiveness grows as the benchmark's T+2 share increases. In Panel B, the discount estimates remain flat and imprecise. This pattern suggests that settlement exposure matters primarily through the asset-gap margin.[24]

[21]The pattern persists when I replace the five-day measure with one-day mispricing split by sign, using $\text{Mispricing}_{f,t-1} \times \mathbf{1}(\text{Mispricing}_{f,t-1} > 0)$ for premiums and $\text{Mispricing}_{f,t-1} \times \mathbf{1}(\text{Mispricing}_{f,t-1} < 0)$ for discounts. The creation-side coefficient remains negative and statistically significant, while the redemption-side coefficient remains statistically indistinguishable from zero.

[22]The April 2024 T+1 settlement booklet published by SIFMA, CCMA and ISDA explains that APs creating US-listed ETFs against foreign securities settling on T+2 or later would need to post collateral for an additional day to receive ETF shares on T+1. J.P. Morgan's T+1 FAQ similarly anticipated larger cash-collateral postings for creation orders.

[23]In a pooled continuous specification, a 10-percentage-point increase in the benchmark's T+2 weight makes the premium-response coefficient 0.0117 percentage points more negative ($p = 0.080$). The estimated premium effect is −0.0147 at the 10th percentile of T+2 weight and −0.1113 at the 90th percentile. The corresponding discount interaction is 0.0026 ($p = 0.761$).

[24]Section 6 shows that the result is robust to alternative sample windows. In the event-time specification, the pre-transition interactions are jointly insignificant for every mispricing measure, while the post-transition interactions are jointly significant for overall mispricing and premiums. Placebo transitions imposed 20 to 80 trading days before the actual transition produce no significant coefficient.

4.2.3 Arbitrage Under Stress: Extreme-Flow States

Balance-sheet and funding constraints should become more binding when aggregate flows increase the resources APs need to complete arbitrage trades. For each benchmark and day, I sum fund-level inflows and outflows separately across all ETFs tracking that benchmark. I define an extreme inflow or outflow day as one on which the corresponding aggregate flow lies in the top decile of that benchmark's own distribution. These states place unusually high simultaneous demands on the APs serving funds tied to the same benchmark.

[Table 4 here]

Table 4 interacts the triple-difference terms with the extreme-flow indicators, using premiums on extreme inflow days and discounts on extreme outflow days. On extreme inflow days, the additional creation-side decline in arbitrage responsiveness is 0.134 percentage points. On extreme outflow days, the additional redemption-side decline is 0.292 percentage points. Both interaction estimates are statistically significant. Adding the corresponding lower-order triple-interaction coefficients from these regressions gives total estimated declines of 0.156 and 0.296 percentage points, respectively. These total declines imply approximately $5.3 million less creation activity per fund-day on extreme inflow days and $10.1 million less redemption activity per fund-day on extreme outflow days, for a one-standard-deviation movement in the respective mispricing measure. Arbitrage responsiveness therefore declines more sharply when aggregate flows place greater demands on AP arbitrage capacity, and the redemption impairment emerges when funding needs are especially high.

4.3 Testing the Mechanism: Authorized Participant Balance-Sheet Constraints

The preceding results suggest that the settlement mismatch constrains AP arbitrage through both asset sourcing and funding. I evaluate this mechanism using two complementary tests. First, I test whether the effect is stronger when the settlement gap spans a weekend. Second, I test whether the effect is stronger among ETFs served by APs with weaker balance sheets. I then examine FX risk as an alternative explanation.

4.3.1 Time-Variation Test: The Thursday Funding Gap

I exploit weekday variation in the duration of the settlement mismatch: for trades initiated on Thursday, the gap spans the weekend. In a redemption, the AP pays for ETF shares on Friday but receives the offsetting cash from the foreign-basket leg only on Monday. The cash-funding gap therefore increases from roughly one calendar day to three. In a creation, the asset gap also extends over the weekend. However, an AP that bridges this gap by posting collateral already faces that requirement on other weekdays. The weekend keeps the existing collateral committed for longer, potentially requiring top-ups, without introducing a new asset-delivery requirement.[25] Proceeds from selling the ETF shares settle on Friday and can help finance this collateral commitment, whereas redemption proceeds remain unavailable until Monday. The weekend extension may therefore make financing constraints more binding for redemption-side arbitrage. I test this prediction by interacting a Thursday indicator with the triple-difference terms.

[Table 5 here]

Panel A of Table 5 shows an additional Thursday decline in redemption-side arbitrage responsiveness of 0.098 percentage points ($p = 0.026$). The corresponding premium interaction is 0.021 ($p = 0.581$). The additional Thursday decline in arbitrage responsiveness is significantly larger for redemptions than for creations.[26] The results distinguish the two margins of the settlement friction. Creation-side arbitrage responsiveness declines on average across the week, consistent with a recurring collateral and asset-delivery constraint. The Thursday estimate does not detect an additional creation-side decline. Redemption-side arbitrage responsiveness declines especially when the cash-funding gap extends across the weekend. The weekday evidence therefore suggests that financing duration becomes particularly costly when APs must bridge a three-day cash shortfall. I next test whether AP balance-sheet capacity moderates these effects directly.

[25] For example, Simplify's Statement of Additional Information requires cash collateral of at least 115% of the value of missing deposit securities, maintained against their daily marked-to-market value until delivery.

[26] The redemption-minus-creation coefficient difference is -0.119 ($p = 0.029$).

4.3.2 Cross-Sectional Test: AP Balance-Sheet Capacity

I classify bank APs as "Weak APs" or "Strong APs" using their Tier 1 capital ratios in 2023, one year before the settlement-cycle change.[27] Weak APs have below-median capital ratios. I then classify an ETF as having strong APs if strong APs account for more than 50% of its 2023 creation and redemption volume among exposed bank APs with available balance-sheet data. The matched APs must cover at least 30% of the fund's total AP activity.

Panel B of Table 5 reports the formal capacity interactions. Greater AP balance-sheet capacity attenuates the creation-side effect by 0.142 percentage points ($p = 0.002$). The corresponding interactions are 0.023 for overall mispricing and 0.022 for discounts; neither is statistically distinguishable from zero. The formal comparison therefore isolates capacity attenuation on the creation side, where the settlement mismatch creates an asset-sourcing gap. Appendix Table C2 reports the corresponding group-specific estimates.

4.3.3 Alternative Explanation: FX Risk

Foreign exchange risk offers an alternative explanation for the reduction in arbitrage responsiveness. US-listed global ETFs tracking European indices face FX risk, whereas European ETFs tracking the same index do not. A change in FX volatility around the settlement-cycle transition could therefore generate an asymmetric response. The transition does not directly change FX settlement: the securities mismatch arises in the Depository Trust & Clearing Corporation (DTCC) system, while FX transactions settle through the Continuous Linked Settlement (CLS) system or bilateral arrangements. Nevertheless, I test the FX explanation using differences in APs' hedging capacity.

Large broker-dealers with substantial FX operations can hedge the currency exposure of global ETF arbitrage at lower cost. If FX risk drives the results, the settlement-mismatch effect should be weaker among APs with greater FX capacity. I classify APs as "High FX Volume" or "Low FX Volume" based on their FX trading activity. High-FX-volume APs are the five largest FX broker-dealers in the Euromoney FX Survey bank-volume ranking. As in the balance-sheet test, I classify an ETF by whether these dealers account for at least 50% of its creation and redemption activity in 2023.

[27] The Tier 1 capital ratio measures a bank's core loss-absorbing capital relative to its risk-weighted assets. For comparable capital requirements, a higher ratio indicates more room to support additional exposures without raising new capital. I therefore use this ratio as a proxy for bank APs' balance-sheet capacity. The Federal Reserve provides the regulatory definition of the Tier 1 capital ratio.

Table C3 presents the results. The triple-interaction coefficient is −0.052 for ETFs primarily served by low-FX-volume APs and −0.074 for those primarily served by high-FX-volume APs. Both estimates are statistically significant at the 1% level. The premium coefficients are also negative and significant in both groups, while the discount coefficients are statistically insignificant. The impairment is therefore not attenuated among ETFs served by APs with large FX operations, contrary to the prediction of the FX-capacity explanation.[28]

4.4 Investor Outcomes: Portfolio Tilting, Tracking Error, and Liquidity

In this section, I examine portfolio tilting, tracking error, and liquidity to assess the consequences of the settlement mismatch for investors.

4.4.1 Portfolio Tilting

I first test whether exposed funds change their holdings. I estimate the difference-in-differences specification of (1) using the portfolio weights allocated to US securities, all T+1-settled securities, and T+2-settled securities as dependent variables.

[Table 6 here]

Panel A of Table 6 shows that exposed funds tilt toward securities whose settlement cycle matches their own.[29] After the transition, the portfolio weight on US securities increases by 0.357 percentage points, and the weight on all T+1-settled securities increases by 0.764 percentage points. The weight on T+2-settled securities falls by 0.609 percentage points. These opposing changes document a shift away from mismatched-settlement securities and toward matched-settlement securities. This adjustment reduces settlement exposure but moves the portfolio away from its foreign benchmark.

4.4.2 Tracking Error

I next examine whether investors receive less precise benchmark tracking. I estimate

[28] A pooled specification formally tests whether the estimates differ between the two groups by interacting the full treatment term with the high-FX-volume indicator and including all lower-order interactions. The high-minus-low-FX differences are −0.028 for signed mispricing, −0.044 for premiums, and −0.004 for discounts, with p-values of 0.359, 0.223, and 0.925, respectively. None is statistically distinguishable from zero.

[29] The portfolio-tilting analysis uses the subset of funds for which holdings data are available.

$$\text{RMS tracking error}_{f,b,t} = \beta\, \mathbf{1}(\text{US}_f) \times \mathbf{1}(\text{Post}_t) + \Gamma \mathbf{X}_{f,b,t-1} + \alpha_p + \alpha_t + \epsilon_{f,b,t}, \tag{3}$$

where RMS tracking error equals $\sqrt{\frac{1}{22}\sum_{i=0}^{21}(R_{1,t-i} - R_{2,t-i})^2}$. The specification includes matched-pair and day fixed effects.

Panel B of Table 6 reports the results. Price-to-benchmark RMS tracking error increases by 0.055 percentage points after the transition. Price-to-NAV tracking error increases by 0.060 percentage points, consistent with weaker price–NAV alignment when primary-market arbitrage declines. NAV-to-benchmark tracking error increases by 0.053 percentage points.[30] The estimated 5.5-basis-point increase in price-to-benchmark RMS tracking error is approximately 11% of the treated funds' pre-transition mean of 0.49% (Table 1). Figure C1 shows the weekly dynamics of tracking-error volatility around the transition.

4.4.3 Liquidity Provision in the Secondary Market

Finally, I examine whether the decline in primary-market arbitrage extends to secondary-market liquidity. I estimate the following difference-in-differences regression:

$$\text{Illiquidity}_{f,b,t} = \beta\, \mathbf{1}(\text{US}_f) \times \mathbf{1}(\text{Post}_t) + \Gamma \mathbf{X}_{f,b,t-1} + \alpha_p + \alpha_t + \epsilon_{f,b,t}, \tag{4}$$

where illiquidity is measured by the Roll (1984) impact ratio or the share of zero-return days. The specification includes matched-pair and day fixed effects.

Panel C of Table 6 shows that liquidity deteriorates after the settlement-cycle change. The coefficient on $\mathbf{1}(\text{US}) \times \mathbf{1}(\text{Post})$ is 0.077 for the Roll impact ratio and 0.003 for the share of zero-return days. Both estimates indicate higher trading frictions. Figure C2 shows a gradual deterioration after the transition, with some fluctuation before the event.

[30] Because the daily RMS measures use overlapping 22-day windows, I also estimate tracking-error volatility at a non-overlapping weekly frequency. For each calendar week, I calculate the standard deviation of active returns and estimate the difference-in-differences specification at the fund-week level. Table C4 reports coefficients of 0.055, 0.056, and 0.051 percentage points for price–benchmark, price–NAV, and NAV–benchmark tracking error, respectively. These estimates closely match the daily RMS estimates. The weekly regressions include matched-pair and week fixed effects and are reported in the appendix.

4.5 Network Propagation: Spillovers to Unexposed Domestic ETFs

The results so far concern funds that face the settlement mismatch directly. I next ask whether the friction propagates through the AP network to US-listed domestic ETFs that share these arbitrageurs but face no settlement mismatch themselves. Both the existence and the direction of such a spillover are open empirical questions.

If APs allocate separate arbitrage capacity to global and domestic ETFs, the shock should remain confined to directly exposed funds. If APs instead reallocate a flexible pool of capital away from newly costly cross-border arbitrage, domestic arbitrage may increase. Conversely, if both activities face a shared AP resource constraint, the settlement mismatch may raise the marginal cost of domestic arbitrage and reduce arbitrage in otherwise unexposed ETFs. Thus, no spillover, a positive spillover, and a negative spillover are all plausible ex ante.

Distinguishing these alternatives requires observing which funds share APs and how those APs respond to the transition. I first characterize the common AP network and document changes in exposed APs' domestic activity. I then construct predetermined fund-level exposure from ETF–AP trading shares and test for spillovers. Finally, I examine whether changes in active relationships, creation activity, and potential substitute capacity offset the propagation.

4.5.1 Characterizing the Global AP–ETF Network

SEC Form N-CEN filings identify the APs that serve each ETF and report the annual value of creation and redemption activity between each fund and AP. I use these data to measure both the links between funds and APs and the economic importance of each link.

[Figure 6]

Figure 6 shows that the AP–ETF network is concentrated and highly interconnected. Panel A maps the APs serving the 10 largest equity ETFs in my sample. These funds trade with 24 APs. Panel B expands the network to the 50 largest equity ETFs, but the number of APs increases only to 27. The network therefore expands mainly through additional ETF links to the existing AP set rather than through the addition of many APs. This structure resembles the dense core documented by Gorbatikov and Sikorskaya (2021).

[Figure 7]

Figure 7 illustrates how APs connect US-listed domestic and global equity ETFs. Blue flows represent creation and redemption activity with domestic equity ETFs, while red flows represent activity with global equity ETFs. Creations appear on the left and redemptions on the right, with wider flows indicating greater trade value. Of the 30 APs with positive equity ETF activity in 2023, 19 serve both ETF types, 10 serve only domestic equity ETFs, and one serves only global equity ETFs. Among APs serving both types, the allocation of activity between domestic and global ETFs varies. Exposed APs collectively handle 96.1% of gross domestic-equity creation and redemption activity. This overlap provides a potential path through which the settlement friction can propagate to domestic ETFs.

4.5.2 Reduced Activity by Exposed APs

I first examine how domestic-equity ETF activity changes around the settlement transition for APs with and without positive pre-transition global ETF activity.[31]

[Table 7 here]

Table 7 shows that exposed APs reduce domestic activity relative to unexposed APs. From 2023 to 2024, their domestic-equity gross activity grows by 0.797 log points less ($p = 0.029$). The annual event-study estimates are small and statistically insignificant in 2021 and 2022, relative to the 2023 base year. Appendix Figure C3 provides a complementary activity decomposition. Domestic creations rise by 9% for exposed APs and 38% for unexposed APs; domestic redemptions rise by 4% and 49%, respectively. Gross activity therefore rises for both groups, but it grows less for APs connected to global ETFs.

4.5.3 The Spillover of the Settlement Friction

I next use one global ETF to illustrate how these links map into fund-level indirect exposure.

[Figure 8]

[31] Appendix Table C5 reports pre-transition AP characteristics. Exposed APs have more active ETF relationships and larger within-ETF activity shares, reflecting the concentration of global ETF activity among major APs.

Figure 8 illustrates the construction using VEA, a US-listed global ETF that invests in developed-market equities outside the United States. Panel A identifies VEA's APs. Panel B shades each domestic-equity ETF by the share of its primary-market trading handled by those APs. The empirical measure extends this mapping to all global ETFs: it identifies APs with pre-reform cross-border activity and weights each domestic ETF's exposure by its 2023 AP trading shares, fixing exposure before the settlement transition.[32]

I use a network shift-share design to test whether settlement pressure propagates to US-listed domestic ETFs that face no settlement mismatch of their own. The exposure combines predetermined ETF–AP trading shares with AP-level cross-border activity. The shares, $s_{f,a}^{pre}$, equal the pre-period fraction of domestic ETF f's primary-market trading conducted with AP a. The AP-level shifter equals one when AP a's pre-period trading volume with US-listed global ETFs exceeds the median across APs. Because more than half of APs report no cross-border activity in 2023, the median is zero; the indicator therefore distinguishes APs with positive global ETF activity from those without it. Before standardization, indirect exposure is:

$$\text{Indirect Exposure}_f = \sum_a s_{f,a}^{pre} \times \text{Exposure}_a^{pre} = \sum_a \frac{\text{Trade}_{f,a}^{pre}}{\text{Trade}_f^{pre}} \times \mathbf{1}(\text{AP}_a^{\text{Exposed}}), \tag{5}$$

where $s_{f,a}^{pre} = \text{Trade}_{f,a}^{pre}/\text{Trade}_f^{pre}$ is AP a's pre-period share of US-listed domestic ETF f's primary-market trading, $\text{Trade}_{f,a}^{pre}$ is the pre-period trading volume between US-listed domestic ETF f and AP a, and Trade_f^{pre} is US-listed domestic ETF f's total pre-period primary-market trading volume. $\mathbf{1}(\text{AP}_a^{\text{Exposed}})$ equals 1 if AP a has positive pre-period trading volume with US-listed global ETFs, and zero otherwise. This share lies between zero and one.[33] Within US-listed domestic ETFs, I estimate the following regression:

[32] Figure 9 recalculates exposure using each year's AP trading shares over 2020–2023, holding the AP classification fixed. Adjacent-year Spearman rank correlations range from 0.49 to 0.56, and 74% to 77% of ETFs remain in the same or an adjacent exposure quintile. These patterns indicate a persistent component of funds' reliance on exposed APs.

[33] Although AP classification is binary, fund-level exposure is continuous because it measures the share of an ETF's pre-transition activity handled by exposed APs. The spillover tests exploit differences in the degree of reliance on these APs. For estimation, I standardize this share to mean zero and unit standard deviation in the regression sample. This metric captures predetermined reliance on APs with pre-period cross-border activity. Because US-listed domestic ETFs do not themselves face a settlement mismatch, variation in the shift-share measure captures indirect exposure through shared arbitrageurs.

$$
\begin{aligned}
\text{Flow}_{f,b,t} = {} & \beta_1 \,\text{Indirect Exposure}_f \times \mathbf{1}(\text{Post}_t) \times \text{Mispricing}_{f,t} \\
& + \beta_2 \,\text{Indirect Exposure}_f \times \mathbf{1}(\text{Post}_t) + \beta_3 \,\mathbf{1}(\text{Post}_t) \times \text{Mispricing}_{f,t} \\
& + \beta_4 \,\text{Indirect Exposure}_f \times \text{Mispricing}_{f,t} + \beta_5 \,\text{Mispricing}_{f,t} \\
& + \Gamma \mathbf{X}_{f,b,t-1} + \alpha_f + \alpha_t + \epsilon_{f,b,t}
\end{aligned}
\tag{6}
$$

The coefficient β_1 compares post-transition changes in arbitrage responsiveness across domestic ETFs with different predetermined reliance on exposed APs. Fund fixed effects absorb persistent fund characteristics, and day fixed effects absorb shocks common to domestic ETFs. Identification requires that, absent the settlement pressure on exposed APs, responsiveness would have evolved similarly across exposure levels.

[Table 8 here]

Table 8 reports the spillover estimates on a common sample of 247 ETFs with available AP-capacity and network measures. Column (1) shows that a one-standard-deviation increase in predetermined indirect exposure reduces post-transition arbitrage responsiveness by 0.042 percentage points ($p = 0.013$).[34] Domestic ETFs more reliant on exposed APs therefore experience a larger decline in arbitrage responsiveness even though they face no settlement mismatch themselves.

4.5.4 AP Capacity and Netting

I next examine whether spillovers depend on AP balance-sheet capacity and opportunities to economize on that capacity through netting. Column (2) shows that AP balance-sheet capacity attenuates the spillover. The estimate is −0.082 for networks with a below-median AP-capacity index. An above-median index adds 0.081 ($p = 0.028$). The index measures the 2023 trading share handled by bank APs in the upper half of the Tier 1 capital-ratio distribution, calculated among exposed bank APs with available data.

Column (3) examines whether spillovers vary with global-equity netting and trading burden, measured using 2023 activity and the same AP trading-share weights, w_{fa}. The network imbalance

[34] In the broader sample, the result remains negative and significant when allowing post-transition arbitrage responsiveness to vary with value-weighted pre-transition AP size (−0.034, $p = 0.046$), or when excluding the globally largest AP (−0.040, $p = 0.002$) or three largest APs (−0.026, $p = 0.044$). The effects associated with each fund's largest AP and its remaining APs do not differ significantly ($p = 0.515$).

ratio is

$$\text{Imbalance}_f = \frac{\left|\sum_a w_{fa}(C_a - R_a)\right|}{\sum_a w_{fa}(C_a + R_a)},$$

where C_a and R_a are AP a's global-equity creation and redemption values. Appendix Figure C4 plots these creation and redemption values by AP. Good netting denotes a ratio at or below the median.[35] High burden denotes networks with both an above-median weighted global-equity trading share and above-median weighted log total activity. Median cutoffs are fixed before restricting the estimation sample.

Better netting attenuates spillovers, especially when APs face a high global-equity trading burden. The good-netting–high-burden interaction is 0.216 ($p < 0.001$). Within high-burden networks, good netting makes the spillover coefficient 0.149 percentage points less negative ($p = 0.007$). This attenuation remains significant after including AP balance-sheet capacity: column (4) reports positive capacity and netting–burden interactions of 0.074 ($p = 0.039$) and 0.150 ($p = 0.012$), respectively. These results suggest that netting helps APs use existing balance-sheet capacity more efficiently, providing protection beyond measured capitalization.[36] I also examine network breadth, concentration, and AP composition in Appendix Table C6, but find little evidence that these characteristics provide additional protection against spillovers.[37] Appendix Table C7 reports alternative specifications using a combined domestic- and global-equity imbalance measure.

4.5.5 Network Adjustment and Partial Replacement

The negative spillover raises the question of whether other APs replace the activity displaced by exposed APs. Domestic ETFs generally authorize several APs. I examine two margins of network adjustment: whether active AP–ETF relationships persist and how creation activity shifts across APs.

[Table 9 here]

Panel A of Table 9 uses an AP–ETF–year panel to follow global ETFs' relationships that were

[35] Opposite AP imbalances can cancel because the absolute value is taken after aggregation. This annual measure captures potential offsetting, not observed settlement-window netting.

[36] Related research explains how offsetting positions and clearing arrangements can reduce intermediaries' funding and collateral needs; see Kirk et al. (2014) and Duffie et al. (2015).

[37] All specifications use the same 247 ETFs and above-median indicators. The standalone active-AP-count interaction is insignificant ($p = 0.260$), while higher AP concentration predicts stronger spillovers ($p = 0.016$). Neither non-bank nor HFT AP shares significantly moderate transmission. When all four characteristics enter jointly with AP capacity, netting, and burden, none of their interactions is statistically significant.

active or registered in 2023 through 2025.[38] The specification compares APs with different exposure levels[39] serving the same ETF over the same reporting period. AP–ETF relationship fixed effects absorb persistent matching, and fund–reporting-period fixed effects absorb fund-level changes over each filing window.[40] Among relationships active in 2023, a one-standard-deviation increase in AP exposure is associated with a 16.90-percentage-point lower probability of remaining active ($p = 0.044$). Among relationships registered in 2023, the corresponding estimate for remaining registered is −7.22 percentage points and is statistically insignificant ($p = 0.143$). Across the sample, 63.67% of active relationships remain active through 2025, while 91.18% of registered relationships remain registered. Exposed APs therefore reduce activity while generally remaining registered to serve the same ETFs.

Panel B uses an ETF–year panel for a difference-in-differences comparison of global and domestic funds. After the transition, global ETFs have 0.924 fewer active APs and a 4.22-percentage-point lower retention rate for active AP relationships, relative to domestic ETFs. The corresponding estimate for retention of registered AP relationships is −0.47 percentage points and is imprecisely estimated. Both designs thus indicate that adjustment occurs mainly in actual AP activity, while registered relationships largely persist.

Panel C asks how much of the creation activity lost by higher-exposure APs is offset by APs with lower global-ETF exposure within the same domestic ETF. For each ETF, I divide lower-exposure APs' increases in creation value by higher-exposure APs' decreases.[41] I compare annual creation values in 2023 and 2025 and repeat the calculation for 2021 and 2023 as a placebo comparison. Among ETFs where higher-exposure APs reduce creations, lower-exposure APs offset a median 27.36% of the lost value. The within-ETF comparison shows that the fraction offset is 14.43 percentage points higher than in the placebo window. After subtracting these increases from higher-exposure APs' decreases, the remaining amount equals a median 10.11% of each ETF's total 2023 creation value.

[38] Registered relationships identify APs authorized to create or redeem shares of an ETF. Active relationships identify registered APs that conducted creation or redemption activity with the ETF during the year. I refer to the APs in the former group as "registered APs" and those in the latter as "active APs".

[39] An AP's exposure is the share of its 2023 gross equity-ETF activity conducted in global ETFs, excluding the focal ETF.

[40] This is the within-ETF comparison in the spirit of Khwaja and Mian (2008).

[41] I cap this fraction at 100%.

4.5.6 Potential Substitute Capacity

Finally, I test whether stronger potential substitutes attenuate the spillover. To measure capacity available beyond an ETF's main existing providers, I exclude its largest active APs when calculating substitute characteristics. These APs already handle much of the ETF's primary-market activity; the question is whether other APs have the capacity to take on additional business. I rank active APs within each domestic ETF by their 2023 gross creation and redemption activity. Column (1) of Table 10 measures the characteristics of the remaining registered APs after excluding the largest active AP; column (2) excludes the three largest. Potential substitutes include both the remaining active APs and registered APs with no creation or redemption activity for the ETF in 2023. The sample contains ETFs with more than five active APs.

[Table 10 here]

I estimate a separate regression for each capacity measure. Each regression augments the spillover specification with

$$\text{Potential Substitute Capacity}_f \times \text{Indirect Exposure}_f \times \mathbf{1}(\text{Post}_t) \times \text{Mispricing}_{f,t}.$$

A positive coefficient indicates that stronger potential substitutes attenuate the exposure-related post-transition decline in the flow–mispricing slope.

Table 10 shows the strongest attenuation for bank size. For a one-standard-deviation increase in indirect exposure, a one-standard-deviation increase in average potential-substitute bank assets attenuates the post-transition decline in the flow–mispricing slope by 0.079 percentage points after excluding the largest active AP and by 0.076 after excluding the three largest active APs. Both estimates are statistically significant at the 1% level. Average AP activity-rank score also predicts attenuation: the corresponding coefficients are 0.073 after excluding the largest active AP and 0.076 after excluding the three largest active APs, statistically significant at the 10% and 5% levels, respectively. The remaining coefficients are 0.043 and 0.080 for the regulatory capital ratio, defined as regulatory capital divided by required minimum capital; 0.041 and 0.035 for better direct equity netting; and 0.051 and 0.030 for the netting share.

5 Interpreting the Evidence: Primary-Market Responsiveness on Shared Balance Sheets

5.1 Model Overview and Empirical Object

The empirical analysis estimates a reduced-form conditional relation between primary-market flows and lagged price–NAV deviations. The model provides an economic interpretation of this relation. Let $\Lambda_f^P \equiv \partial X_f^P / \partial m_f$ denote the local response of aggregate primary-market positions to the mispricing signal in the model, holding the other state variables fixed. The model derives qualitative comparative statics that can be compared with the sign and cross-sectional variation of the estimated flow–mispricing coefficient. Secondary-market arbitrage affects the equilibrium level of mispricing but does not change shares outstanding. I therefore model primary-market responsiveness in the text and provide the secondary-market closure in Appendix E.

The framework builds on the segmented-market logic of Gromb and Vayanos (2002), as adapted to ETF primary markets by Gorbatikov and Sikorskaya (2021). In the Gorbatikov and Sikorskaya benchmark, proportional intermediation costs affect participation but not the responsiveness of a continuing AP. The shared convex cost introduced here illustrates an intensive-margin mechanism through which a continuing AP can become less responsive when settlement pressure raises the marginal cost of its common balance sheet. The baseline model develops this mechanism for one AP serving a global and a domestic ETF. The extension aggregates heterogeneous APs and maps their individual responses into the shift-share exposure measure.

5.2 Baseline Model: One AP Serving Two ETF Pairs

5.2.1 Environment and the AP's Problem

There are two dates. Investors and arbitrageurs trade at date 1, and assets pay at date 2. There are two segmented ETF–basket pairs, indexed by $f \in \{X, D\}$. Pair X consists of a US-listed global ETF and its foreign basket. Pair D consists of a US-listed domestic ETF and its domestic basket. Within each pair, the ETF and basket are claims to the same date-2 payoff. Their payoffs and investor demands are otherwise unrelated.

Each market contains a unit mass of price-taking CARA investors with risk aversion γ_f. Investors can trade only the claim in their own market and receive opposite endowment exposures to the common payoff. Let u_f denote the resulting date-1 demand imbalance. As in Gorbatikov and Sikorskaya (2021), these assumptions produce linear demand.

Let X_f^P denote the aggregate primary-market AP position and let X_f^S denote the position of secondary-market arbitrageurs. Total arbitrage is $Z_f = X_f^P + X_f^S$. Market clearing gives

$$m_f = 2b_f(u_f - Z_f), \qquad b_f \equiv \gamma_f \sigma_f^2, \qquad \frac{1}{2}\left(p_f^{\text{basket}} + p_f^{\text{ETF}}\right) = \bar{p}_f, \tag{7}$$

where m_f is the price–NAV deviation that arbitrage closes, σ_f^2 is the variance of the common payoff, b_f measures inverse market depth, and $\bar{p}_f$ is the pair's average price. One unit of arbitrage raises the price of the cheap claim by b_f, lowers the price of the expensive claim by b_f, and closes the deviation by $2b_f$. An investor-demand shock therefore opens a deviation, while primary- and secondary-market arbitrage close it. The empirical counterpart of m_f is the price–NAV deviation constructed in Appendix D.

Consider one risk-neutral AP, indexed by a, with predetermined access to both pairs. The AP chooses primary-market positions x_{aX} and x_{aD}. Its aggregate balance-sheet usage is

$$\Gamma_a = \bar{\Lambda}_a + \varphi_{aX} x_{aX} + \varphi_{aD} x_{aD}, \tag{8}$$

where $\varphi_{af} > 0$ is the capacity consumed per unit of arbitrage and $\bar{\Lambda}_a$ is balance-sheet usage predetermined at the daily horizon. It includes unsettled obligations from earlier trades and other uses of the AP's balance sheet; it is not capital committed to continue arbitraging. The AP remains free to reduce current positions x_{af}, including to zero. The AP pays the shared cost $\Psi_a(\Gamma_a)$. I impose

$$\Psi_a'(\Gamma_a) > 0, \qquad \Psi_a''(\Gamma_a) > 0, \qquad \Psi_a'''(\Gamma_a) > 0 \quad \text{locally around the relevant equilibrium.} \tag{9}$$

The first two conditions say that using more balance sheet is costly and that the cost of the next trade rises with usage. The third says that this increase becomes sharper as capacity fills: an extra position matters little when the AP has ample room, but much more when little room remains. A hard regulatory or internal limit can produce this pattern, but the model does not require the limit

to bind. This cost shape is motivated by the elastic marginal-cost cases in Eisenbach and Phelan (2026), who analyze power costs $C(q) = cq^n$ with $n > 2$, and by Buchak et al. (2024), whose marginal portfolio-funding cost steepens as a bank's capital buffer approaches its regulatory minimum. Because Ψ_a depends on aggregate usage, positions in the two pairs are linked through

$$\frac{\partial^2 \Psi_a(\Gamma_a)}{\partial x_{aX} \partial x_{aD}} = \varphi_{aX} \varphi_{aD} \Psi_a''(\Gamma_a) > 0. \tag{10}$$

The AP solves

$$\max_{x_{aX}, x_{aD} \geq 0} \left\{ \sum_{f \in \{X,D\}} \left[x_{af} m_f\left(x_{af}, x_{-a,f}; X_f^S\right) - 2\bar{p}_f C_{af} x_{af} \right] - \Psi_a(\Gamma_a) \right\}, \tag{11}$$

where $x_{-a,f}$ denotes other AP positions and equals zero in the one-AP baseline. Competition is Cournot within each pair. When choosing x_{af}, the AP treats other primary- and secondary-market positions as fixed and recognizes that $\partial m_f / \partial x_{af} = -2b_f$. The term C_{af} collects pair-specific execution, borrowing, and creation or redemption costs. The two pairs meet only through the shared balance-sheet cost.

5.2.2 Settlement Mismatch and Shared Balance-Sheet Usage

The US transition to T+1 creates the settlement bridge

$$h_t = \text{Post}_t \, w^{T+2} \Delta T, \tag{12}$$

where w^{T+2} is the basket weight that remains on T+2 and ΔT is the settlement gap. In the one-AP baseline, the bridge raises the capacity required per unit of cross-border arbitrage,

$$\varphi_{aX} = \varphi_{aX}^0 + \gamma_\varphi h_t, \qquad \gamma_\varphi > 0, \tag{13}$$

and raises the short-run balance-sheet load from unsettled obligations. Because APs trade repeatedly, the settlement mismatch creates a rolling pipeline. At the daily decision horizon, financing, collateral, or delivery obligations generated by earlier trades may remain outstanding when the AP chooses its current positions. Opposite-direction obligations may offset, but imperfect netting leaves a residual balance-sheet load. Let Q_a^+ and Q_a^- denote opposite-direction cross-border obligations over the

settlement window. Define gross pipeline scale, netting inefficiency, and residual unsettled obligations by

$$F_a = Q_a^+ + Q_a^-, \qquad 1 - n_a = \frac{|Q_a^+ - Q_a^-|}{Q_a^+ + Q_a^-}, \qquad R_a = F_a(1 - n_a). \tag{14}$$

Predetermined usage is

$$\bar{\Lambda}_a = A_a + \gamma_\Lambda h_t R_a, \qquad \gamma_\Lambda > 0. \tag{15}$$

Here A_a is the AP's baseline predetermined usage. Thus, φ_{aX} captures the capacity consumed by a new cross-border trade, whereas $\bar{\Lambda}_a$ captures balance-sheet usage inherited at the daily decision horizon. Poor netting and large gross scale therefore reinforce one another:

$$\frac{\partial^2 \bar{\Lambda}_a}{\partial F_a \partial (1 - n_a)} = \gamma_\Lambda h_t > 0. \tag{16}$$

The bridge is a reduced-form shifter of balance-sheet usage.

5.2.3 Direct and Spillover Effects

Using $\partial m_f / \partial x_{af} = -2b_f$, the first-order conditions for an interior solution are

$$m_f = 2b_f x_{af} + 2\bar{p}_f C_{af} + \varphi_{af} \Psi_a'(\Gamma_a), \qquad f \in \{X, D\}. \tag{17}$$

Let $\omega_a = \Psi_a''(\Gamma_a)$ and define

$$\kappa_{af} = \frac{2b_{-f}}{2b_{-f} + \varphi_{a,-f}^2 \omega_a} \in (0, 1). \tag{18}$$

Here $-f$ denotes the other ETF–basket pair. The AP's responsiveness in pair f, holding the other pair's signal fixed, is

$$\lambda_{af} \equiv \frac{\partial x_{af}}{\partial m_f} = \frac{1}{2b_f + \varphi_{af}^2 \omega_a \kappa_{af}}. \tag{19}$$

Proposition 1 (Direct effect)**.** *A wider settlement bridge has a negative mechanical effect on cross-border responsiveness through φ_{aX}, holding aggregate usage fixed. The total effect is*

$$\frac{d\lambda_{aX}}{dh_t} = -\lambda_{aX}^2 \left[2\gamma_\varphi \varphi_{aX} \omega_a \kappa_{aX} + \varphi_{aX}^2 \kappa_{aX}^2 \Psi_a'''(\Gamma_a) \frac{d\Gamma_a}{dh_t} \right], \tag{20}$$

which is negative whenever the bracket is positive.

The first term is the direct settlement-intensity effect. The second captures the movement in marginal-cost sensitivity as aggregate balance-sheet usage changes.

Proposition 2 (Spillover to the domestic pair)**.** *The domestic pair has no settlement mismatch of its own. Its responsiveness changes according to*

$$\frac{d\lambda_{aD}}{dh_t} = -\lambda_{aD}^2 \varphi_{aD}^2 \kappa_{aD}^2 \left[\Psi_a'''(\Gamma_a) \frac{d\Gamma_a}{dh_t} - \frac{\gamma_\varphi \varphi_{aX} \omega_a^2}{b_X} \right]. \tag{21}$$

The spillover is negative if and only if the expression in brackets is positive.

The two terms in (21) work in opposite directions. Greater aggregate usage moves the AP to a region where marginal balance-sheet cost is more sensitive to additional activity. This curvature channel lowers domestic responsiveness. At the same time, the AP reduces newly expensive cross-border arbitrage and releases discretionary capacity to the domestic book. This reallocation force raises domestic responsiveness. The reduction in current arbitrage does not immediately eliminate the predetermined settlement load in $\bar{\Lambda}_a$. A negative spillover requires aggregate usage to rise and the curvature channel to dominate the released-capacity effect. In particular, aggregate usage rises when

$$\gamma_\Lambda R_a + \gamma_\varphi x_{aX} > -\varphi_{aX} \frac{dx_{aX}}{dh_t} - \varphi_{aD} \frac{dx_{aD}}{dh_t}. \tag{22}$$

The observed negative spillover selects this regime within the model.

5.3 Extension to Heterogeneous APs

5.3.1 AP Exposure and Aggregate Responsiveness

Let $\mathcal{A}_f$ be the APs with predetermined access to pair f, and let $z_a \in [0,1]$ measure AP a's predetermined exposure to cross-border settlement activity. The AP-specific bridge is

$$h_{at} = z_a h_t. \tag{23}$$

The one-AP results apply to each AP after replacing h_t with h_{at}. The empirical exposed-AP indicator E_a is a coarse classification of this underlying exposure.

Aggregate domestic primary-market responsiveness is

$$\Lambda_D^P = \sum_{a \in \mathcal{A}_D} \lambda_{aD}. \tag{24}$$

When exposed APs become less responsive, lower-exposure APs may conduct more domestic arbitrage because the equilibrium price–NAV deviation becomes larger. This quantity response moves alternative APs along their existing supply schedules. It does not necessarily make those schedules more responsive to a given deviation. Aggregate conditional responsiveness therefore falls unless the slopes of alternative APs rise enough to offset the exposed APs' slope loss. Appendix E provides the formal replacement and participation decompositions.

5.3.2 Mapping the Model to the Shift-Share Design

Let $\sigma_{Da}^{pre} = \lambda_{aD,pre}/\Lambda_{D,pre}^P$ denote AP a's pre-period share of domestic responsiveness, and let $\tau_a = -\Delta\lambda_{aD}/\lambda_{aD,pre}$ denote its proportional slope loss. Then

$$\frac{\Lambda_{D,post}^P - \Lambda_{D,pre}^P}{\Lambda_{D,pre}^P} = -\sum_{a \in \mathcal{A}_D} \sigma_{Da}^{pre} \tau_a \simeq -\tau \sum_{a \in \mathcal{A}_D} \sigma_{Da}^{pre} E_a \simeq -\tau \text{IndExp}_D. \tag{25}$$

The first approximation holds when slope losses concentrate among high-exposure APs and are similar within that group, with τ denoting their common proportional slope loss. The second uses pre-period AP activity shares to proxy unobserved slope shares. Under these approximations, the model maps the negative-spillover regime into a negative reduced-form coefficient on $\text{IndExp}_D \times \text{Post}_t \times m_{Dt}$ in (6).

5.4 Empirical Predictions and Scope

The framework yields four predictions that map to the empirical analysis. First, the settlement bridge lowers cross-border responsiveness, and the decline grows with the basket weight in T+2-settled securities. Second, the friction propagates to domestic ETFs in proportion to their predetermined reliance on exposed APs when the curvature channel dominates capacity release and replacement. Third, greater AP balance-sheet capacity attenuates both effects, holding usage and other primitives fixed, in the power and shrinking-buffer cost specifications reported in Appendix E.

Fourth, poor netting and large gross global scale jointly raise residual unsettled usage through (16). These predictions correspond to the direct-effect, AP-capacity, and network-exposure tests in Section 4.2, Section 4.3, and Section 4.5.

Because secondary-market arbitrage also affects equilibrium price deviations, the framework makes predictions for primary-market responsiveness rather than unconditional average mispricing. The appendix provides the secondary-market closure, the creation–redemption extension, the nesting of Gorbatikov and Sikorskaya (2021), and the full derivations.

6 Additional Identification and Validation Tests

I examine the timing and sample stability of the direct creation-side result, test placebo transition dates, and consider the 2017 settlement transition.

Event-time dynamics and sample stability. Appendix Table C8 replaces the single post indicator with 21-trading-day event periods over a symmetric $[-90, +90]$ window and omits the month before the transition. For creation-side arbitrage, the pre-transition coefficients are jointly insignificant ($p = 0.958$), while the post-transition coefficients are jointly significant ($p < 0.001$). The first two post-transition coefficients are -0.082 and -0.173, although the estimates vary across later periods.

The creation-side result is also stable across sample windows. The premium coefficient remains between -0.080 and -0.098 and statistically significant at the 1% level when I exclude the transition days, the August 2024 volatility episode, or the fourth quarter of 2024, or restrict the sample to a symmetric $[-60, +60]$ window. The pooled mispricing coefficient is less stable: it is -0.030 ($p = 0.105$) when excluding the fourth quarter and -0.041 ($p = 0.101$) in the $[-60, +60]$ window.

Placebo transition dates. Appendix Table C9 re-estimates the baseline using placebo transition dates 80, 60, 40, and 20 trading days before 28 May 2024. Each regression uses only pre-transition observations. None of the eight triple-interaction coefficients is statistically significant, and seven have the opposite sign from the corresponding baseline estimate. The placebo tests do not reproduce the baseline decline.

Historical audit: The 2017 T+3-to-T+2 transition. The 2017 US transition from T+3 to T+2 narrowed the settlement mismatch between US-listed global ETF shares and foreign securities already settling on T+2. The settlement-friction mechanism therefore predicts an increase in arbitrage responsiveness. Appendix Table C10 reports four positive estimates, ranging from 0.036 to 0.048. Their signs are consistent with this prediction, but none is statistically significant.[42]

7 Conclusion

This paper examines how a local operational constraint spreads across distinct arbitrage trades through shared arbitrageurs. The 2024 US transition to T+1 raised the resources required to arbitrage global ETFs whose foreign baskets continued to settle on T+2. Primary-market arbitrage responsiveness declines for directly exposed ETFs relative to non-US ETFs tracking the same benchmarks. Creation-side arbitrage responsiveness declines on average, while redemption impairment emerges when the funding gap spans a weekend or aggregate outflows are extreme. Greater AP balance-sheet capacity attenuates the creation-side impairment. These patterns are consistent with asset-sourcing and funding needs constraining arbitrage. Funds tilt toward securities with matching settlement cycles, but tracking error rises and secondary-market liquidity deteriorates.

The constraint also propagates to domestic ETFs through shared APs, even though their shares and underlying baskets settle on the same cycle. Greater AP balance-sheet capacity attenuates these spillovers. Better netting also attenuates transmission, especially when APs face a high global-equity trading burden, suggesting that propagation depends on both available capacity and how efficiently it is used. Other APs offset part of the creation activity lost by higher-exposure APs, and stronger potential substitutes predict smaller spillovers. These findings show that the reach of an arbitrage constraint depends on how capacity is shared across trades and on other arbitrageurs' ability to absorb displaced activity.

These findings raise a broader policy question: when should rules governing connected markets be harmonized, and when should they remain locally tailored? Harmonization can reduce operational frictions across markets, but may constrain jurisdictions' ability to pursue domestic objectives. Locally tailored rules preserve regulatory autonomy, yet differences in rules can create artificial segmentation

[42] The exercise uses the 2024 benchmark mapping and funds observed in both periods.

that impairs trading across connected markets. The settlement mismatch studied here illustrates this cost. This tradeoff is particularly consequential when otherwise distinct markets depend on shared intermediaries with limited capacity. Future research can examine when the benefits of cross-market coordination outweigh those of local flexibility, and how the availability and organization of intermediation shape that balance.

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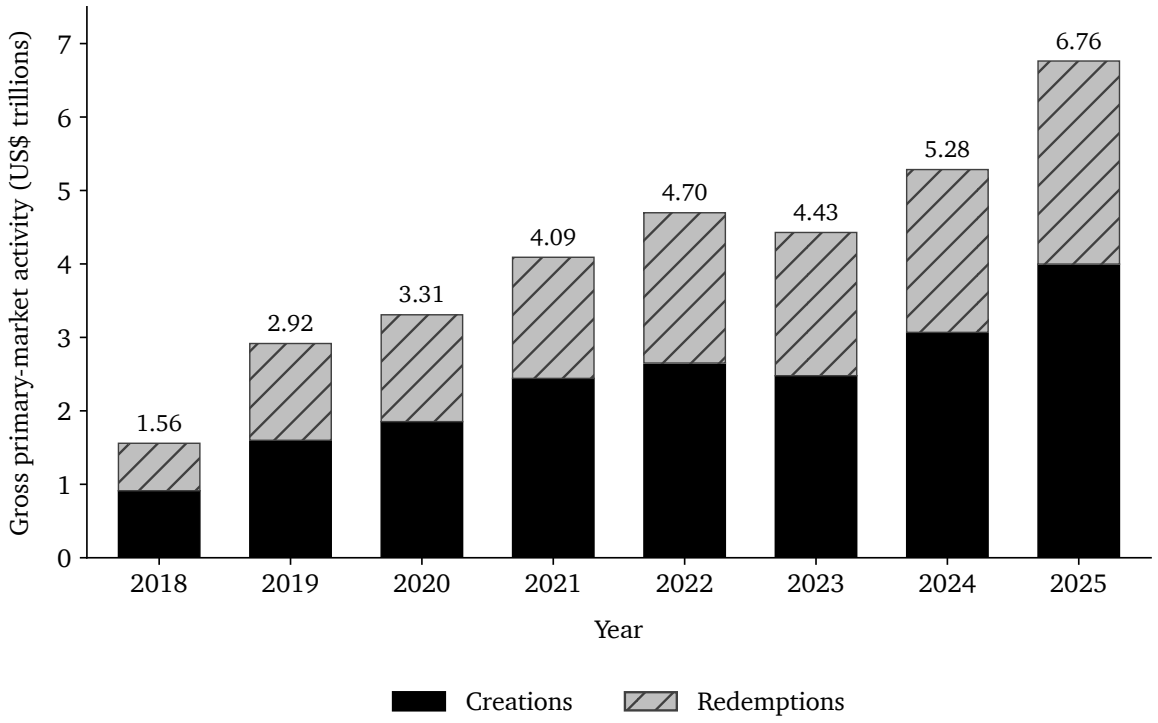
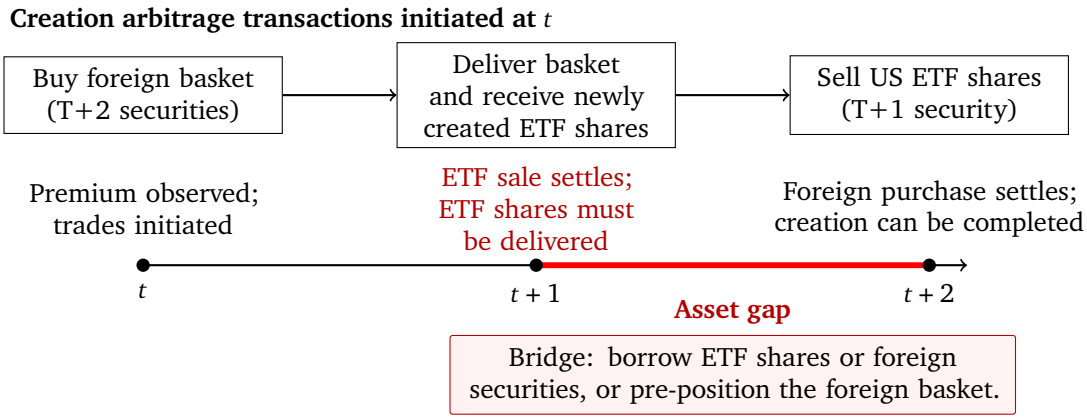
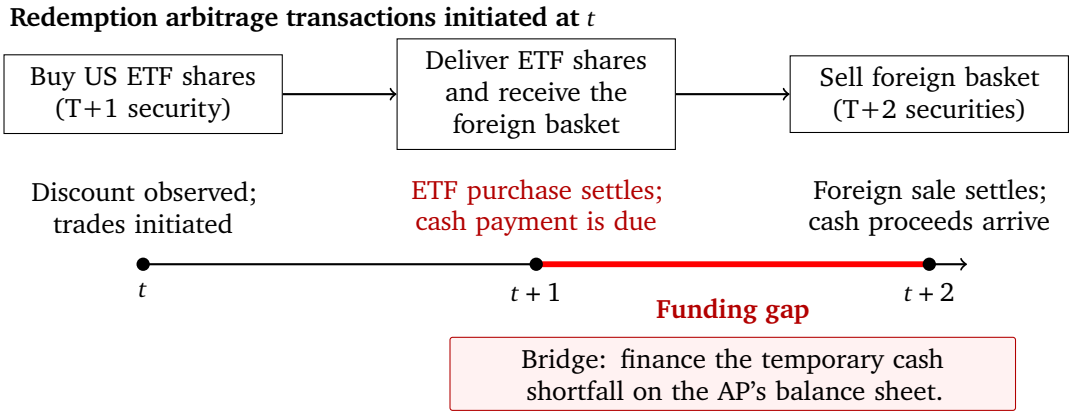


Figure 1. Annual ETF Primary-Market Activity

This figure reports gross ETF creations and redemptions from SEC Form N-CEN for fiscal periods ending in 2018–2025, in trillions of US dollars. Black bars denote creations, and hatched gray bars denote redemptions. Labels above the bars report their sum. Amounts are aggregated across APs and ETFs. The sample includes all ETFs with reported transaction amounts.



(a) Creation arbitrage when the ETF trades at a premium: the asset gap



(b) Redemption arbitrage when the ETF trades at a discount: the funding gap

Figure 2. Settlement Mismatch in Global ETF Arbitrage

This figure illustrates settlement mismatch for US-listed ETF shares settling on T+1 and foreign basket securities settling on T+2. Panel A shows creation arbitrage: ETF shares must be delivered before the foreign-basket purchase settles. Panel B shows redemption arbitrage: payment for ETF shares precedes the proceeds from selling the foreign basket. Red intervals mark the one-day asset and funding gaps.

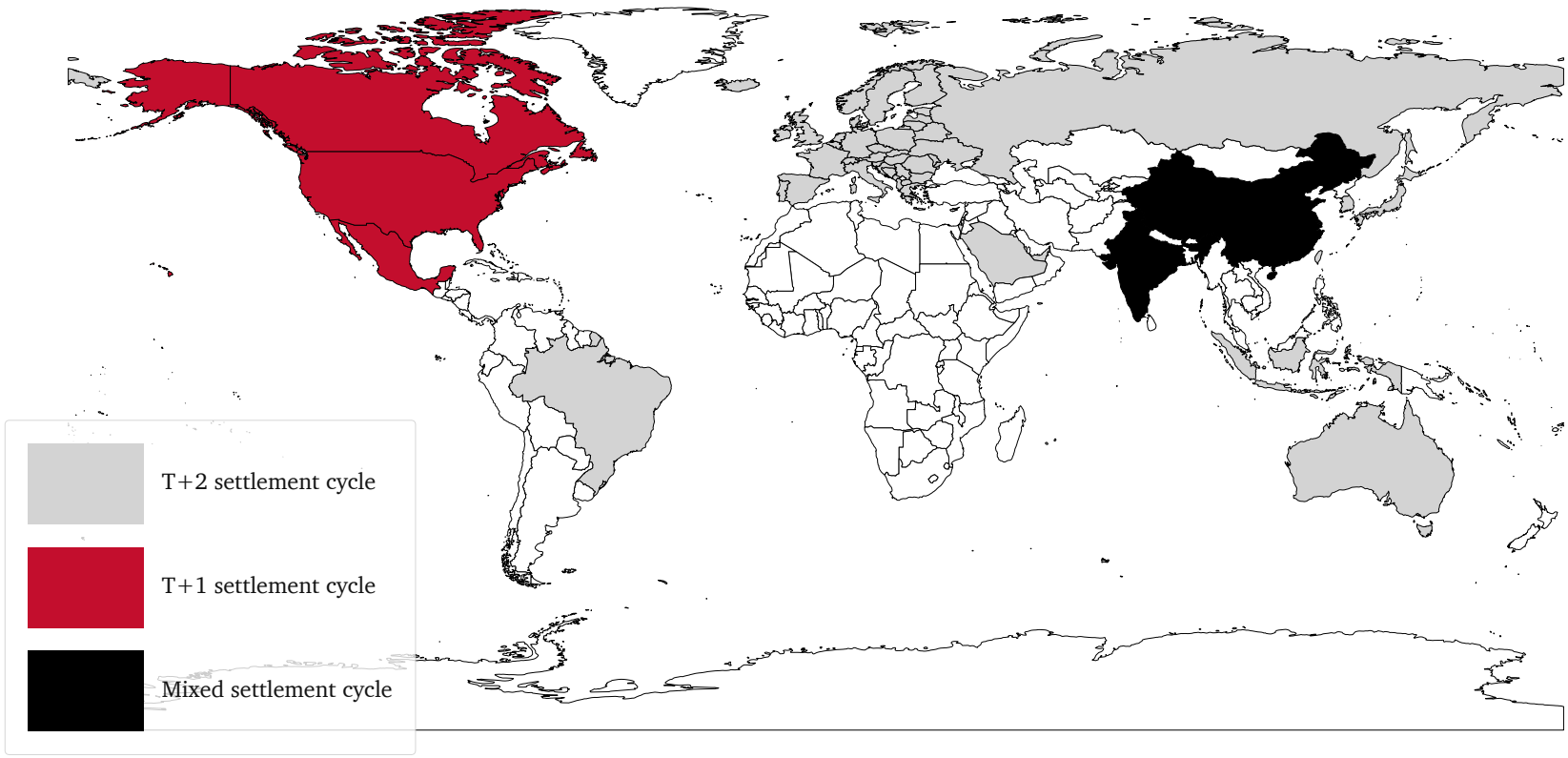


Figure 3. Settlement Cycles across Major Jurisdictions

This figure shows settlement cycles for exchange-traded securities in major jurisdictions as of December 2024. Red denotes T+1, gray T+2, and black mixed settlement cycles. Unclassified jurisdictions are unshaded.

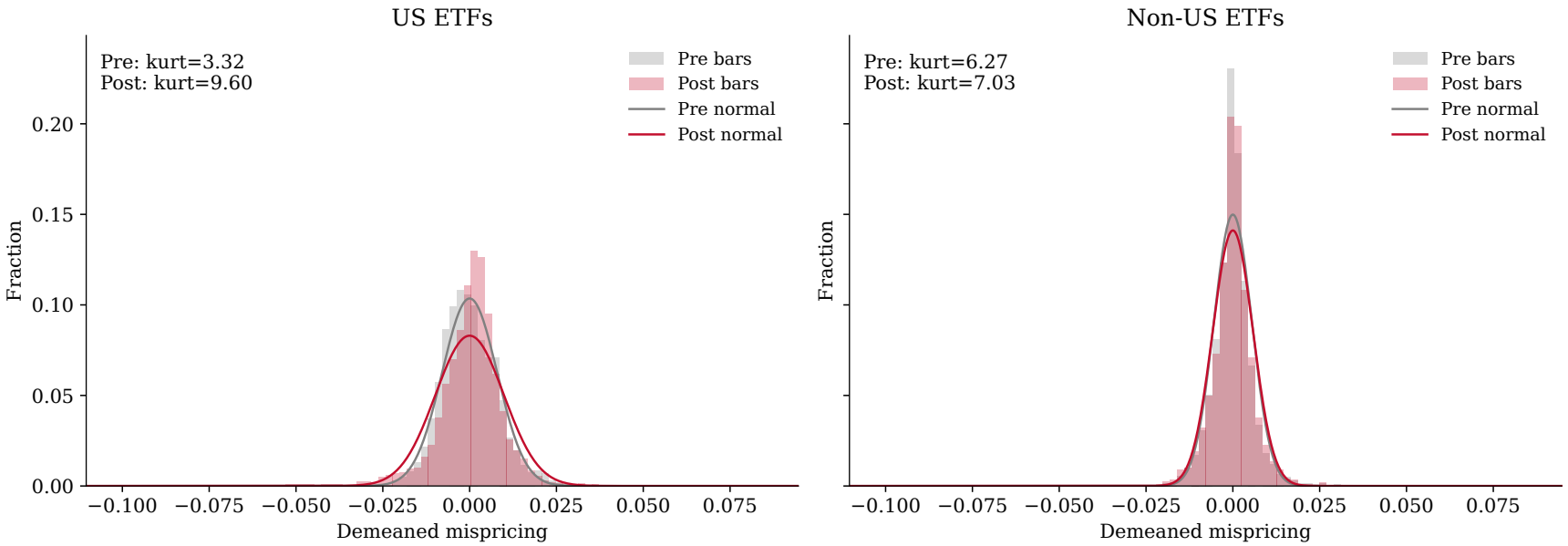
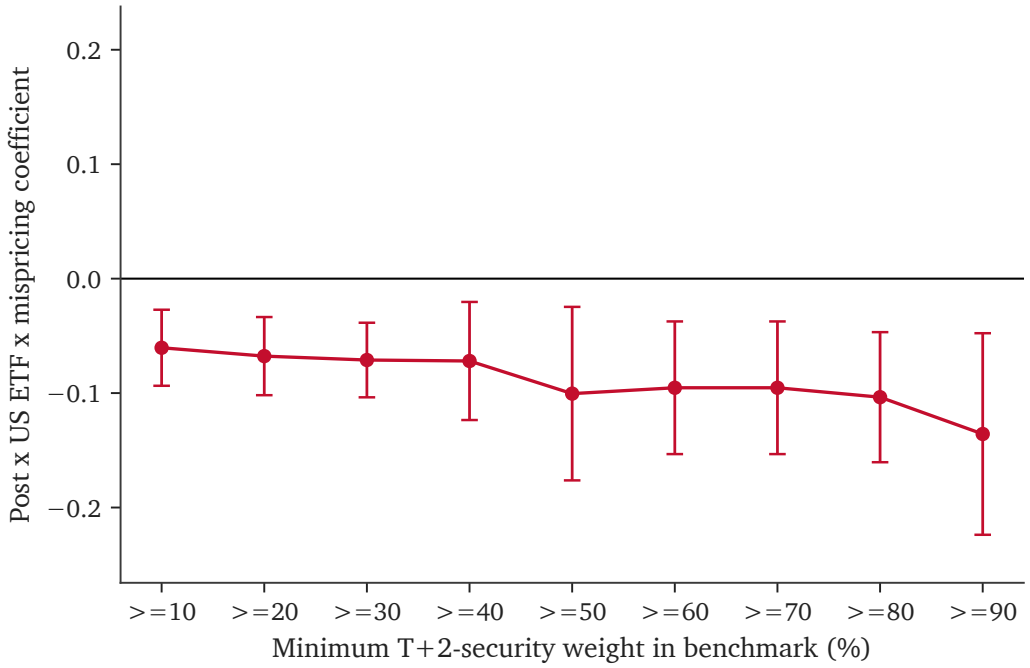
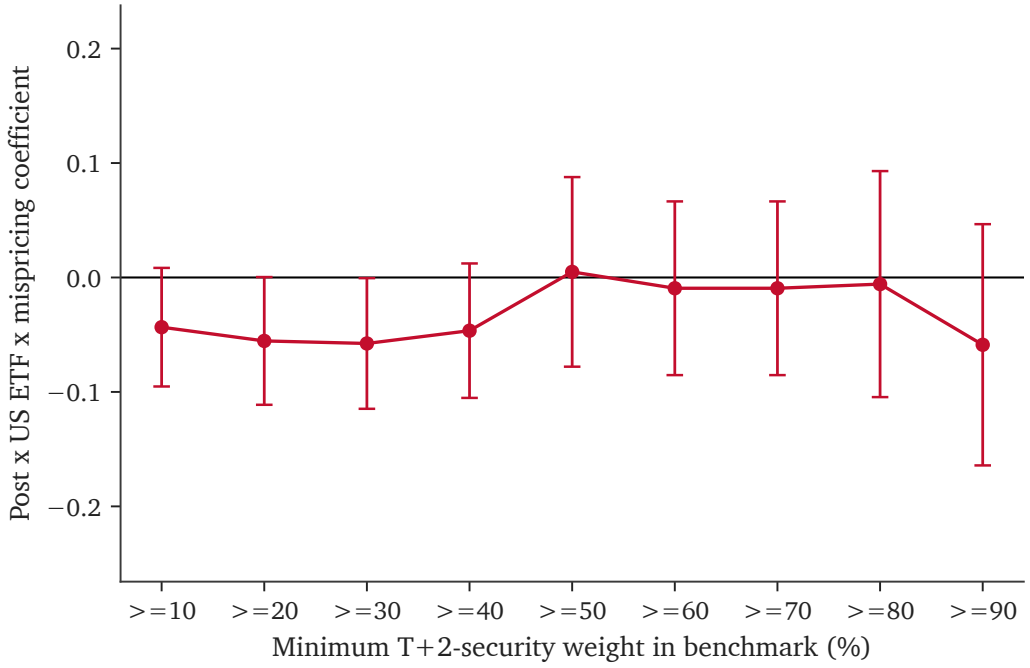


Figure 4. Synchronized Mispricing Distributions Before and After T+1

This figure compares mispricing for US-listed global ETFs on the left and non-US ETFs tracking the same indices on the right. US mispricing uses synchronized prices and matched-fund NAVs; non-US mispricing uses local-close prices and NAVs. Gray denotes observations before May 28, 2024, and red denotes observations on or after that date. Each group-period mean is subtracted. Histograms report observation fractions, and curves show fitted normal distributions. Mispricing is in decimal units; 0.001 equals 10 basis points.



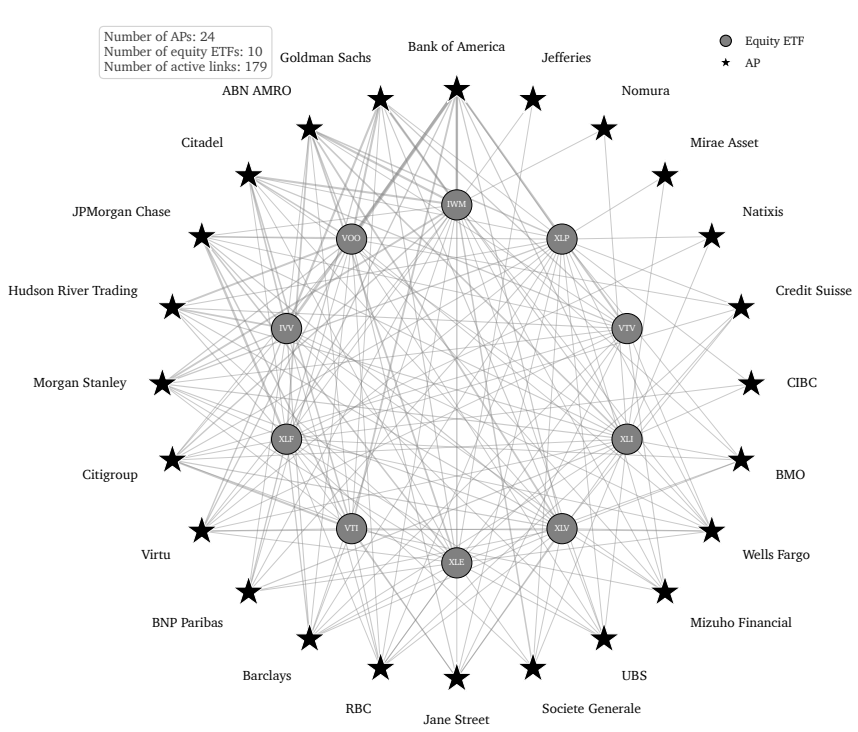
(a) Positive mispricing



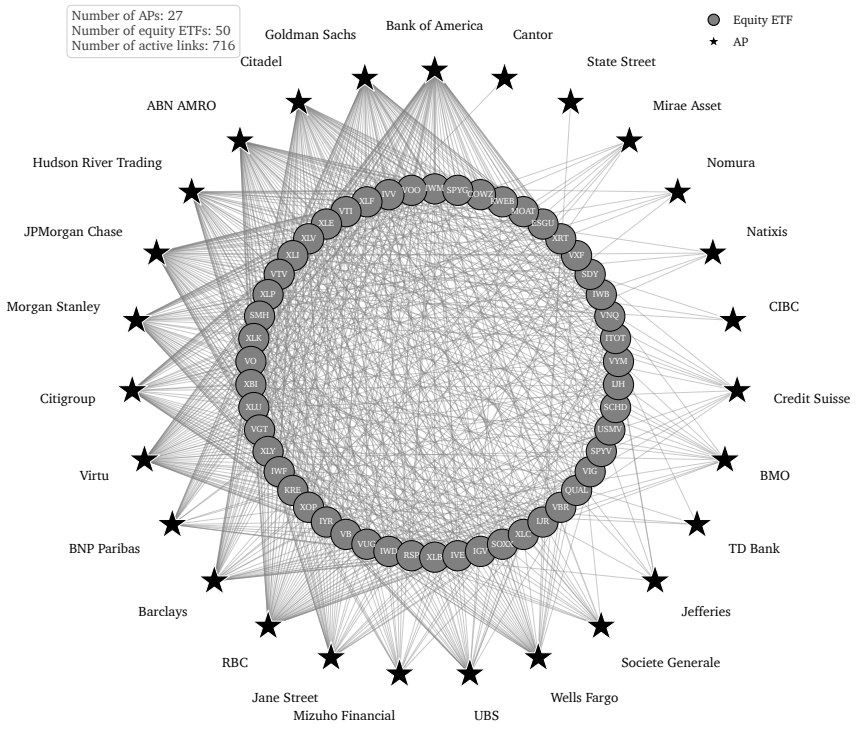
(b) Negative mispricing

Figure 5. Arbitrage Responsiveness by Benchmark Exposure to T+2 Securities

This figure plots the estimated change in arbitrage responsiveness for US-listed global ETFs relative to matched non-US ETFs as the minimum benchmark weight in T+2 securities increases. Each point comes from a separate regression using ETFs whose benchmark T+2 weight is at least the threshold on the horizontal axis. Panel A uses standardized one-day lagged premiums, and Panel B uses standardized one-day lagged discounts. Each measure is zero on the opposite side of price–NAV parity. Vertical bars show 95% confidence intervals.



(a) Ten largest equity ETFs



(b) Fifty largest equity ETFs

Figure 6. Primary-Market Network of Authorized Participants and Equity ETFs

This figure shows AP–ETF trading networks from 2023 SEC Form N-CEN filings. Gray circles represent equity ETFs, and black stars represent APs. Links indicate positive creation or redemption activity; widths are proportional to the square root of gross trade value. Panels A and B show successively broader groups of ETFs ranked by gross primary-market activity.

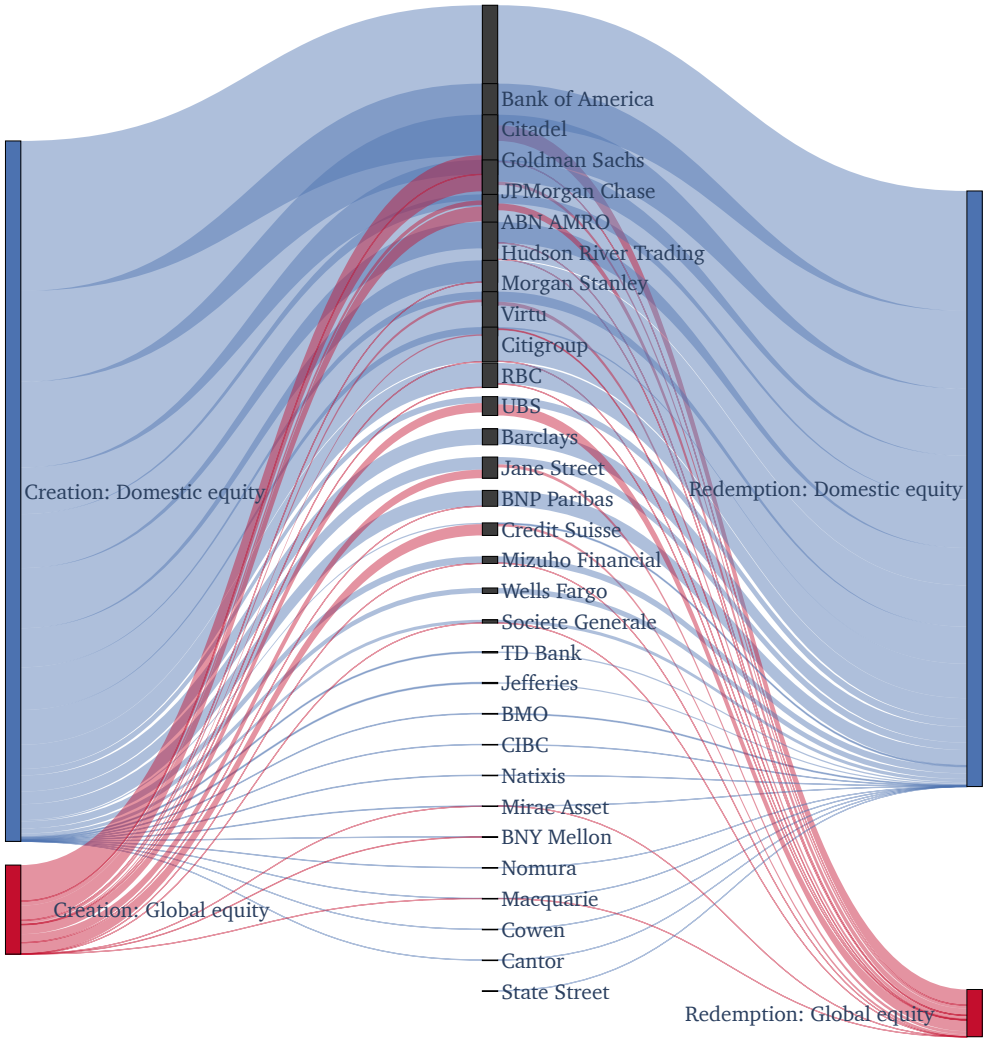
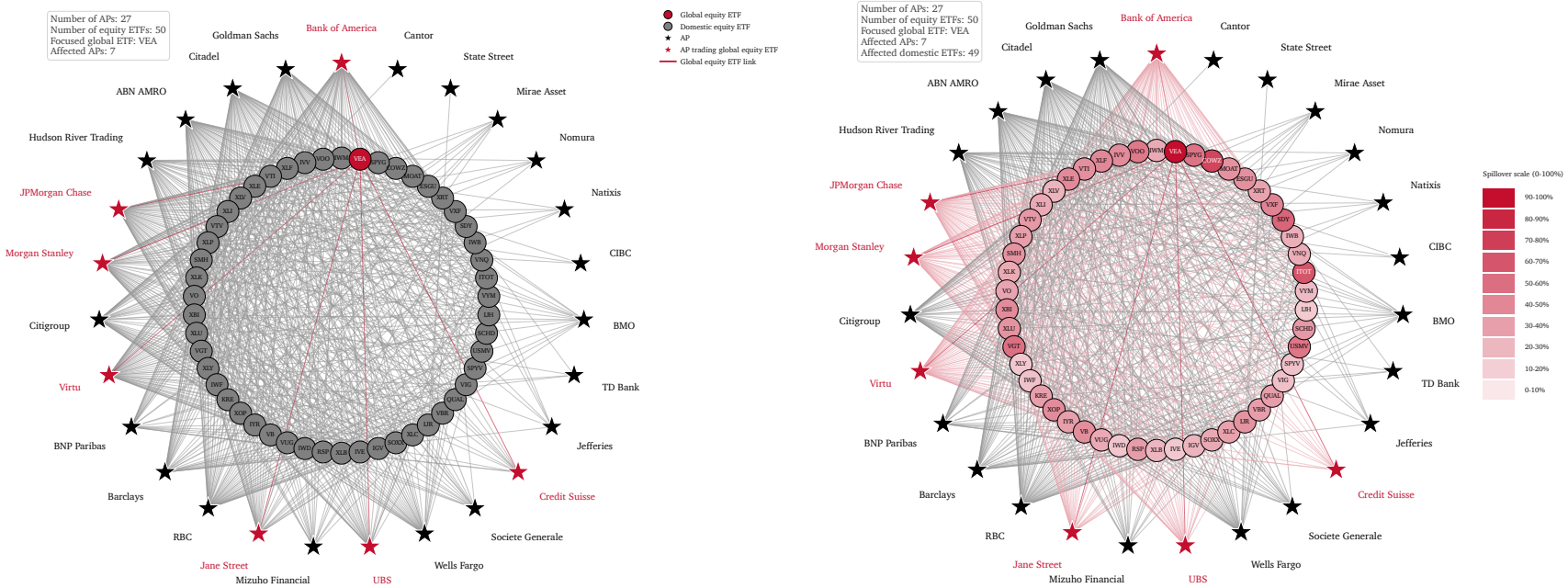


Figure 7. Authorized Participant Activity Across Domestic and Global Equity ETFs

This figure shows AP creation and redemption activity in US-listed domestic and global equity ETFs in 2023, using SEC Form N-CEN. Creations appear on the left, APs in the middle, and redemptions on the right. Blue denotes domestic equity ETFs, and red denotes global equity ETFs. Flow widths represent trade value in billions of US dollars. APs with positive activity in either group are included.



(a) Global links through APs

(b) Domestic ETFs sharing VEA's APs

Figure 8. Authorized Participant Network Channels for Global and Domestic Equity ETFs

This figure shows links between VEA and domestic equity ETFs through shared APs in the large-ETF network in 2023. The network layout and link widths follow Figure 6. Panel A highlights VEA's APs and their links in red. Panel B shades domestic ETFs by the share of their primary-market activity handled by these APs; darker red denotes greater reliance. Data are from SEC Form N-CEN.

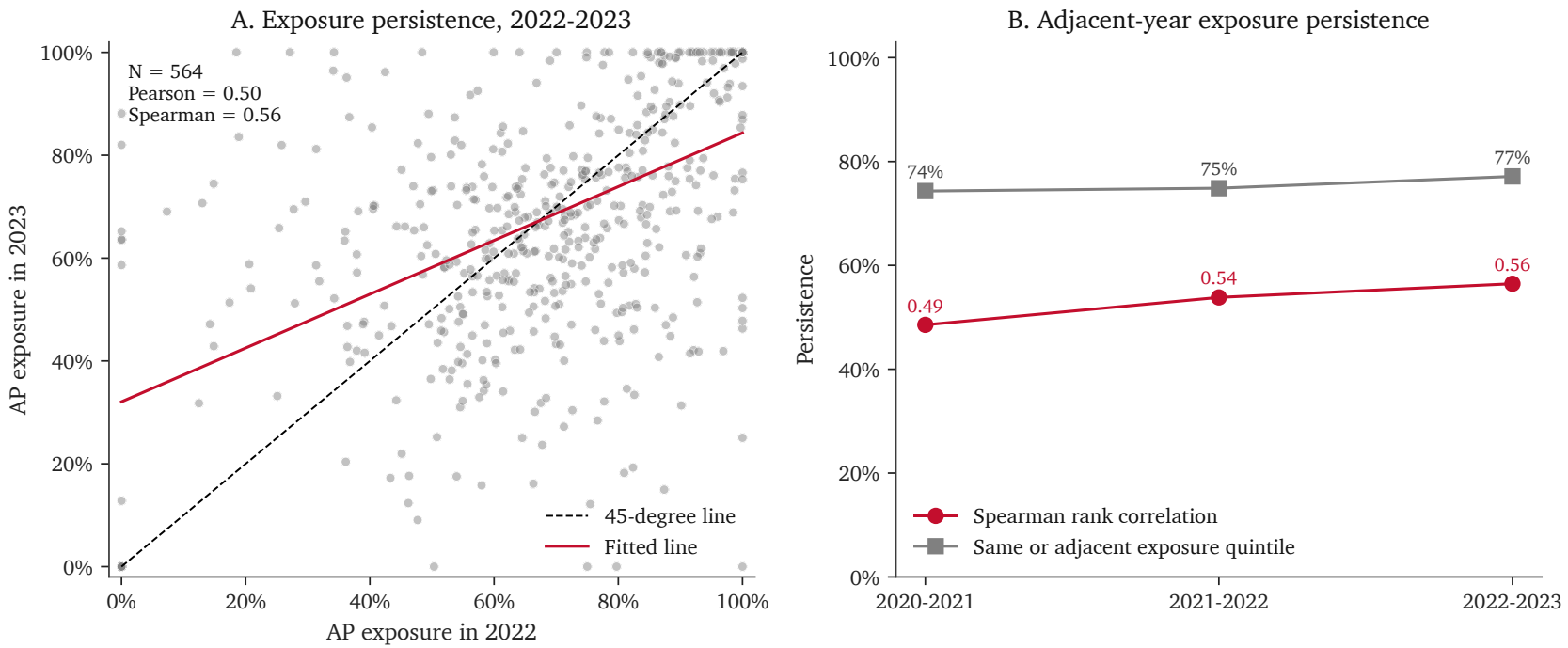


Figure 9. Pre-Reform Persistence of Authorized-Participant Exposure

This figure shows persistence in domestic ETFs' exposure to APs with positive global ETF activity in 2023. Exposure is the share of primary-market trading handled by these APs. AP classification is fixed, while annual ETF–AP trading shares vary. Panel A plots 2022 exposure against 2023 exposure; the dashed line is the 45-degree line, and the solid line is a fitted regression. Panel B reports adjacent-year rank correlations and the share remaining in the same or an adjacent exposure quintile over 2020–2023.

Table 1. Descriptive Statistics of ETFs before the United States T+1 Settlement Cycle Change

This table reports pre-transition ETF characteristics in 2024 by listing market and benchmark exposure. Panel A covers the full sample; Panel B covers ETFs sharing a benchmark across US and non-US listing markets. TNA is total net assets in billions; fund age is years since inception; and expense ratio is the annual net expense ratio in percent. Change in shares outstanding is the signed daily percentage change in the share count. Raw return is the fund's daily reported return. Mispricing is the price–NAV deviation in basis points. RMS tracking error is the root mean squared difference between fund and benchmark returns over 22 trading days. Roll is the Roll (1984) illiquidity measure. Columns report means, standard deviations, and medians.

Panel A: All ETFs

	A. US Benchmark						B. Non-US Benchmark					
	i. US ETFs			ii. Non-US ETFs			i. US ETFs			ii. Non-US ETFs		
	Mean	SD	P50	Mean	SD	P50	Mean	SD	P50	Mean	SD	P50
TNA (Billions)	3.62	7.52	0.43	1.10	3.29	0.18	1.60	4.73	0.17	0.67	2.11	0.08
Fund Age (Years)	9.82	7.04	7.87	5.28	4.19	4.30	9.22	6.21	8.30	5.48	4.66	4.11
Expense Ratio (%)	0.32	0.20	0.30	0.23	0.14	0.20	0.43	0.20	0.45	0.37	0.16	0.35
Change in shares outstanding (%)	0.04	0.88	0.00	0.07	0.94	0.00	0.02	0.73	0.00	0.09	0.88	0.00
Raw Return (%)	0.07	0.96	0.10	0.08	0.93	0.08	0.05	0.97	0.09	0.06	0.97	0.09
Mispricing (bps)	0.71	12.37	0.43	-5.89	92.97	-1.79	1.01	40.69	0.22	-12.11	120.21	1.46
RMS Tracking Error (%)	0.14	0.26	0.07	0.78	0.37	0.72	0.50	0.30	0.48	0.60	0.35	0.55
Roll	0.57	1.23	0.32	0.37	0.86	0.15	0.35	1.34	0.23	0.21	0.32	0.12
Number of Funds		711			279			575			555	

Panel B: ETFs Sharing a Benchmark across Listing Markets

	A. US Benchmark						B. Non-US Benchmark					
	i. US ETFs			ii. Non-US ETFs			i. US ETFs			ii. Non-US ETFs		
	Mean	SD	P50	Mean	SD	P50	Mean	SD	P50	Mean	SD	P50
TNA (Billions)	10.36	11.89	3.10	2.17	5.19	0.33	3.01	6.09	0.39	1.92	4.75	0.20
Fund Age (Years)	12.53	7.52	13.38	6.69	4.79	5.87	11.00	5.40	12.23	8.13	5.41	6.46
Expense Ratio (%)	0.22	0.19	0.16	0.21	0.12	0.20	0.46	0.22	0.49	0.35	0.18	0.20
Change in shares outstanding (%)	0.04	0.72	0.00	0.08	0.85	0.00	-0.02	0.58	-0.00	0.09	0.79	0.00
Raw Return (%)	0.07	0.97	0.10	0.08	0.90	0.10	0.05	1.04	0.11	0.05	0.93	0.11
Mispricing (bps)	-0.71	11.72	0.00	1.60	63.98	-2.70	-3.66	35.07	-4.36	-14.78	132.97	0.54
RMS Tracking Error (%)	0.13	0.24	0.06	0.78	0.28	0.74	0.49	0.25	0.48	0.61	0.25	0.56
Roll	0.93	1.95	0.41	0.59	1.26	0.21	0.37	0.24	0.30	0.25	0.29	0.15
Number of Funds		55			93			33			70	

Table 2. List of Authorized Participants

This table lists AP parent groups with positive activity in the matched US-listed ETF sample during 2018–2024. Trade value is cumulative creations plus redemptions, in millions of US dollars. Domestic and global ETF shares report the allocation of each AP's activity by benchmark exposure. The sample includes equity, bond, and other ETFs.

Authorized Participant	Institution Type	Trade Value (USD Mn)	Domestic ETF Share (%)	Global ETF Share (%)
Bank of America	Bank	5,650,234.16	89%	11%
JPMorgan Chase	Bank	3,422,254.67	90%	10%
Goldman Sachs	Bank	3,222,496.94	95%	5%
Citadel	Non-Bank	1,420,969.70	98%	2%
Morgan Stanley	Bank	1,452,543.03	93%	7%
ABN AMRO	Bank	1,388,564.90	97%	3%
Virtu	Non-Bank	1,048,203.73	94%	6%
Citigroup	Bank	845,477.75	97%	3%
Hudson River Trading	Non-Bank	725,494.60	94%	6%
RBC	Bank	633,693.67	96%	4%
BNP Paribas	Bank	469,311.85	97%	3%
Barclays	Bank	372,920.41	99%	1%
Jane Street	Non-Bank	455,473.76	77%	23%
Credit Suisse	Bank	645,547.17	39%	61%
Jefferies	Non-Bank	214,812.44	100%	0%
UBS	Bank	448,839.35	39%	61%
Societe Generale	Bank	206,855.69	83%	17%
Deutsche Bank	Bank	160,275.69	98%	2%
Wells Fargo	Bank	129,986.54	98%	2%
Mizuho Financial	Bank	97,320.79	99%	1%
BNY Mellon	Bank	44,334.11	99%	1%
Cantor	Non-Bank	35,406.24	94%	6%
BMO	Bank	23,037.23	100%	0%
TD Bank	Bank	16,937.12	100%	0%
Natixis	Non-Bank	16,169.16	95%	5%
Nomura	Non-Bank	14,851.52	100%	0%
HSBC	Bank	12,636.81	100%	0%
CIBC	Bank	10,062.40	100%	0%
Cowen	Non-Bank	6,201.51	93%	7%
Macquarie	Non-Bank	4,239.46	74%	26%
Mirae Asset	Non-Bank	2,792.97	88%	12%
Wedbush	Non-Bank	2,142.99	100%	0%
State Street	Bank	1,888.84	100%	0%
Interactive Brokers	Non-Bank	1,042.80	92%	8%
Fidelity	Non-Bank	692.44	100%	0%
Robert W. Baird	Non-Bank	22.20	100%	0%
Clear Street	Non-Bank	4.51	100%	0%
PEAK6 Investments	Non-Bank	2.14	100%	0%
ING	Bank	2.64	0%	100%

Table 3. Primary-Market Arbitrage Responsiveness

This table estimates equation (2) for US-listed global ETFs and non-US ETFs tracking the same benchmark around May 28, 2024. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest is US × Post × Mispricing, where US identifies US-listed funds and Post identifies observations on or after the transition. Controls include one-day lags of ln(TNA), the natural log of total net assets; ln(Fund age), the natural log of years since inception; fund returns, net-of-fee daily returns; flow, the signed daily percentage change in ETF shares outstanding; and Roll, the Roll (1984) impact ratio measuring ETF illiquidity. Columns (1)–(3) use standardized measures of lagged signed mispricing, the five-day maximum premium, and the five-day minimum discount. Regressions include all lower-order interactions and matched-pair and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)		
	Mispricing	Mispricing^{+}	Mispricing^{-}
	(1)	(2)	(3)
1(US) × **1**(Post) × Mispricing	-0.047***	-0.083***	-0.006
	(0.016)	(0.018)	(0.023)
1(Post) × Mispricing	0.016	0.040***	-0.014
	(0.013)	(0.014)	(0.018)
1(US) × Mispricing	0.048***	0.052***	0.033*
	(0.012)	(0.014)	(0.019)
1(US) × **1**(Post)	-0.002	0.005	-0.005
	(0.018)	(0.019)	(0.019)
1(US)	-0.056***	-0.071***	-0.044***
	(0.014)	(0.015)	(0.014)
ln(TNA)	0.009**	0.008*	0.012***
	(0.005)	(0.004)	(0.004)
ln(Fund age)	-0.053***	-0.052***	-0.053***
	(0.013)	(0.013)	(0.013)
Fund returns	0.008	0.008	0.008
	(0.006)	(0.006)	(0.006)
Lagged flow	0.029**	0.028**	0.028**
	(0.012)	(0.012)	(0.012)
Mispricing	-0.013	-0.005	0.011
	(0.011)	(0.011)	(0.016)
Roll	0.012	0.012	0.009
	(0.022)	(0.020)	(0.021)
Matched-pair FE	✓	✓	✓
Day FE	✓	✓	✓
N	34,709	35,677	35,677
R^2	0.0551	0.0557	0.0555

Table 4. Arbitrage Responsiveness under Extreme Benchmark Flows

This table examines arbitrage responsiveness on extreme benchmark-flow days. The dependent variables are net creations in column (1) and net redemptions in column (2), measured as directional daily percentage changes in shares outstanding; redemptions retain their negative sign. The variable of interest interacts US, Post, an extreme-flow indicator, and standardized one-day lagged premiums or discounts. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Lagged mispricing and extreme-flow indicators are also included. Extreme inflows and outflows are the top decile of each benchmark's respective aggregate-flow distribution. Regressions include all lower-order interactions and matched-pair and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Directional change in shares outstanding (%)	
	Creation flow	Redemption flow
	(1)	(2)
$\mathbf{1}(\text{US}) \times \mathbf{1}(\text{Post}) \times \mathbf{1}(\text{Extreme benchmark flow}) \times \text{Mispricing}$	-0.134**	-0.292***
	(0.061)	(0.107)
Controls	✓	✓
Lower-order interactions	✓	✓
Matched-pair FE	✓	✓
Day FE	✓	✓
N	27,850	27,850
R^2	0.1062	0.2337

Table 5. Primary-Market Arbitrage and Authorized Participant Constraints

This table examines how the settlement effect varies with funding duration and AP balance-sheet capacity. The dependent variable is the signed daily percentage change in shares outstanding. The variables of interest interact US × Post × Mispricing with a Thursday indicator or strong AP capacity. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Panel A examines Thursday trades; Panel B examines AP capacity. Strong APs have above-median 2023 Tier 1 capital ratios. An ETF is classified as strong if these APs handle more than 50% of its activity among exposed bank APs with capital data; matched APs must cover at least 30% of total activity. Mispricing measures follow Table 3. Regressions include all lower-order interactions and matched-pair and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: Thursday funding gap

	Dependent variable: Change in shares outstanding (%)		
	Mispricing	Mispricing$^+$	Mispricing$^-$
	(1)	(2)	(3)
$\mathbf{1}$(US) × $\mathbf{1}$(Post) × $\mathbf{1}$(Thursday) × Mispricing	0.007	0.021	-0.098**
	(0.035)	(0.038)	(0.043)
Controls	✓	✓	✓
Lower-order interactions	✓	✓	✓
Matched-pair FE	✓	✓	✓
Day FE	✓	✓	✓
N	34,709	35,677	35,677
R^2	0.0554	0.0559	0.0559

Panel B: AP balance-sheet capacity

	Dependent variable: Change in shares outstanding (%)		
	Mispricing	Mispricing$^+$	Mispricing$^-$
	(4)	(5)	(6)
$\mathbf{1}$(US) × $\mathbf{1}$(Post) × Mispricing × $\mathbf{1}$(Strong AP)	0.023	0.142***	0.022
	(0.032)	(0.045)	(0.050)
Controls	✓	✓	✓
Lower-order interactions	✓	✓	✓
Matched-pair FE	✓	✓	✓
Day FE	✓	✓	✓
N	33,442	34,409	34,409
R^2	0.0579	0.0585	0.0595

Table 6. Settlement-Cycle Mismatch and Investor Outcomes

This table compares changes in investor outcomes for US-listed global ETFs and non-US ETFs tracking the same benchmark. The dependent variables measure portfolio composition, tracking error, and illiquidity. The variable of interest is US × Post. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Panels A and B also control for lagged mispricing; Panel C excludes Roll. Panel A reports portfolio weights in US, T+1, and T+2 securities. Panel B reports 22-trading-day RMS tracking errors between price, NAV, and benchmark returns. Panel C reports Roll illiquidity and the share of zero-return days. Regressions include matched-pair and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: Portfolio tilting

	Dependent variable: Total portfolio weight (%)		
	US securities	T+1 securities	T+2 securities
	(1)	(2)	(3)
$\mathbf{1}(\text{US}) \times \mathbf{1}(\text{Post})$	0.357***	0.764***	-0.609***
	(0.121)	(0.168)	(0.168)
Matched-pair FE	✓	✓	✓
Day FE	✓	✓	✓
Controls	✓	✓	✓
N	9,524	9,524	9,524
R^2	0.9963	0.9855	0.9850

Panel B: RMS tracking error

	Dependent variable: RMS tracking error (%)		
	Price–Benchmark	Price–NAV	NAV–Benchmark
	(4)	(5)	(6)
$\mathbf{1}(\text{US}) \times \mathbf{1}(\text{Post})$	0.055***	0.060***	0.053***
	(0.007)	(0.008)	(0.005)
Matched-pair FE	✓	✓	✓
Day FE	✓	✓	✓
Controls	✓	✓	✓
N	23,524	23,479	23,522
R^2	0.6192	0.6089	0.6729

Panel C: Liquidity provision

	Dependent variable: Illiquidity measures	
	Roll impact ratio	Zero-return days
	(7)	(8)
$\mathbf{1}(\text{US}) \times \mathbf{1}(\text{Post})$	0.077***	0.003***
	(0.013)	(0.000)
Matched-pair FE	✓	✓
Day FE	✓	✓
Controls	✓	✓
N	24,359	37,265
R^2	0.5518	0.0669

Table 7. Changes in Domestic-Equity Activity by Authorized-Participant Exposure

This table compares domestic-equity activity of APs with and without positive global-equity activity in 2023, using SEC Form N-CEN. Gross activity is creations plus redemptions. The dependent variable is the 2023–2024 change in log activity in column (1) and annual log activity in column (2). The variable of interest is exposed-AP status in column (1) and its interactions with year indicators in column (2). Column (2) covers 2021–2024 and omits 2023. Column (1) has no fixed effects; column (2) includes AP and year fixed effects. Standard errors are heteroskedasticity-robust in column (1), clustered by AP in column (2), and reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Domestic-equity gross AP activity	
	(1)	(2)
	$\Delta \ln$(Gross trade)	ln(Gross trade)
	2023→2024	2021–2024
1(Exposed AP)	-0.797**	
	(0.347)	
1(Exposed AP) × 2021		-0.145
		(0.482)
1(Exposed AP) × 2022		0.152
		(0.466)
1(Exposed AP) × 2024		-0.797**
		(0.351)
AP FE		✓
Year FE		✓
N	30	119
R^2	0.1890	0.9207

Table 8. Spillovers, AP Capacity, and Global-Equity Netting

This table examines spillovers to US-listed domestic ETFs. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest is Indirect Exposure × Post × Mispricing and its interactions with AP characteristics. Indirect exposure is standardized pre-transition reliance on APs with positive global ETF activity; mispricing is standardized signed mispricing. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Columns (1)–(4) report the baseline, AP-capacity, netting–burden, and joint specifications. High capacity denotes an above-median trading share handled by upper-half Tier 1 bank APs among exposed bank APs with capital data. Good netting denotes a ratio of absolute weighted net creations to weighted gross activity in global equity ETFs at or below the panel median. The absolute value is taken after aggregation across APs. High burden requires above-median weighted global-equity trading share and total log activity. AP measures use 2023 activity. Regressions include all lower-order interactions and fund and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)			
	Baseline	AP capacity	Netting and burden	Joint
	(1)	(2)	(3)	(4)
Indirect exposure × Post × Mispricing	-0.042**	-0.082***	-0.003	-0.029
	(0.017)	(0.026)	(0.017)	(0.026)
High AP capacity × Indirect exposure × Post × Mispricing		0.081**		0.074**
		(0.036)		(0.036)
Good netting × Indirect exposure × Post × Mispricing			-0.067*	-0.066
			(0.040)	(0.040)
High burden × Indirect exposure × Post × Mispricing			-0.107***	-0.087**
			(0.034)	(0.038)
Good netting × High burden × Indirect exposure × Post × Mispricing			0.216***	0.150**
			(0.056)	(0.059)
Good versus poor netting within high burden			0.149***	0.084
			(0.054)	(0.054)
ETFs	247	247	247	247
Controls	✓	✓	✓	✓
Lower-order interactions	✓	✓	✓	✓
Fund and day FE	✓	✓	✓	✓
N	53,965	53,965	53,965	53,965
R^2	0.0597	0.0608	0.0604	0.0611

Table 9. Authorized-Participant Network Adjustment and Creation Replacement

This table examines AP relationship retention and changes in creation activity using annual SEC Form N-CEN records. The dependent variables are relationship retention in Panel A, active-AP counts and relationship retention in Panel B, and the fraction of lost creations offset and the remaining loss in Panel C. The variables of interest are AP exposure interacted with post-transition reporting share in Panel A, global-ETF status interacted with post-transition reporting share in Panel B, and the policy-window indicator in Panel C. Registered APs are authorized to create or redeem ETF shares; active APs report positive transactions. Panel A follows global ETFs' 2023 relationships through 2025. Exposure is the standardized share of an AP's 2023 equity-ETF activity in global ETFs, excluding the focal fund; post-transition share is the fraction of the reporting window after May 28, 2024. Panel B compares global and domestic ETFs; retention effects are in percentage points. Panel C compares domestic ETFs over 2023–2025 and 2021–2023. The fraction offset divides lower-exposure APs' creation gains by higher-exposure APs' losses, capped at 100%. The remaining loss, floored at zero, is expressed relative to initial total creations. Column (1) reports policy-window medians; column (2) reports policy-minus-placebo estimates. Panel A includes AP–ETF and fund–reporting-period fixed effects; Panel B includes fund and quarter fixed effects; Panel C regressions include fund fixed effects. Standard errors are clustered by AP and fund in Panel A and by fund in Panels B and C, and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Panel A: Within-ETF AP–ETF link retention

	Dependent variable: AP–ETF relationship status	
	1(AP active)	**1**(AP registered)
	(1)	(2)
AP exposure × Post share	-16.90**	-7.22
	(7.79)	(4.80)
2023–2025 retention rate	63.67%	91.18%
AP–ETF relationship FE	✓	✓
Fund × reporting-period FE	✓	✓
N	3,975	21,103
R^2	0.7037	0.5852

Panel B: ETF-level AP-network adjustment

	Dependent variable: ETF-level AP-network outcome		
	Number of active APs	Active-link retention rate	Registered-link retention rate
	(1)	(2)	(3)
Global × Post share	-0.924***	-4.22**	-0.47
	(0.119)	(1.93)	(0.90)
Fund FE	✓	✓	✓
Quarter FE	✓	✓	✓
N	5,653	4,358	4,422
R^2	0.9001	0.4235	0.4262

Panel C: Creation-activity replacement in domestic ETFs

	Dependent variable: Creation-activity replacement measure	
	Policy-window median	Policy minus placebo
	(1)	(2)
Creation decreases offset (%)	27.36%	14.43***
		(4.17)
R^2	—	0.4522
Remaining decrease / initial creations (%)	10.11%	-11.12***
		(2.35)
R^2	—	0.4572

Table 10. Potential-Substitute Characteristics and Spillover Attenuation

This table examines how potential substitute APs moderate spillovers to domestic equity ETFs. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest interacts each standardized substitute characteristic with Indirect Exposure, Post, and signed mispricing; each row reports a separate regression. Exposure and mispricing follow Table 8. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. The remaining registered-AP count is also included. Columns (1) and (2) exclude each ETF's largest and three largest active APs, respectively. Remaining registered APs, including inactive APs, form the substitute pool and receive equal weights. Higher activity rank denotes a stronger trading position; RCR is regulatory capital relative to its required minimum; better direct equity netting denotes lower creation–redemption imbalance relative to gross activity. Eligibility requires more than five active APs in 2023. Regressions include all lower-order interactions and fund and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)	
	Excluding largest active AP	Excluding three largest active APs
	(1)	(2)
Each row reports a separate regression		
AP activity rank		
Average AP activity-rank score	0.073*	0.076**
	(0.043)	(0.037)
Balance-sheet capacity		
ln(Bank total assets)	0.079***	0.076***
	(0.024)	(0.026)
ln(RCR)	0.043*	0.080**
	(0.026)	(0.038)
Balance-sheet efficiency		
Better direct equity netting	0.041	0.035*
	(0.025)	(0.018)
Netting share	0.051***	0.030*
	(0.018)	(0.017)
Included in every regression		
Controls	✓	✓
Lower-order interactions	✓	✓
Number of remaining registered APs	✓	✓
Fund FE	✓	✓
Day FE	✓	✓
N	48,939	48,939

Appendix

A Global arbitrage under settlement cycle mismatch

To illustrate the mechanism, consider a physically replicating US-listed global ETF whose shares settle on T+1 while the foreign securities in its creation and redemption basket settle on T+2. Let t denote the trade date. The same one-day mismatch creates different obligations depending on the direction of arbitrage. Figure 2 maps the transaction and settlement sequence for creations and redemptions.

In creation arbitrage, an ETF premium induces the AP to sell ETF shares and purchase the foreign basket. The ETF sale settles on T+1, but the foreign securities purchased to complete the in-kind creation do not settle until T+2. The AP therefore may have to deliver ETF shares before the basket is available and the creation can be completed. As Section 4.2 describes, bridging this asset gap requires the AP to borrow the ETF shares, to settle against cash collateral where the trust permits it, or to source the foreign securities directly. Each route expands the AP's gross balance sheet: borrowed positions and posted collateral enter the leverage-ratio exposure measure without offsetting the obligation they secure, and required margin in excess of that obligation immobilizes liquid assets for the duration of the bridge.

In redemption arbitrage, an ETF discount induces the AP to purchase ETF shares and submit them for in-kind redemption while selling or short-selling the corresponding foreign basket. The ETF purchase requires a cash payment on T+1, but the foreign-security sale does not generate cash proceeds until T+2. The AP must finance this temporary cash shortfall, so the settlement mismatch consumes funding and balance-sheet capacity. Thus, the same one-day mismatch creates a securities-delivery problem in creations and a cash-financing problem in redemptions. When the mismatch spans a weekend, the creation-side collateral remains committed for longer, while the redemption-side cash-funding gap extends from roughly one day to three calendar days. The Thursday test in Section 4.3.1 compares these responses.

B Sample Selection and Regression Samples

Table B1. Sample Construction and Authorized-Participant Risk Sets

This table summarizes sample construction. Panel A reports the full ETF universe and the same-benchmark matched samples. Panel B applies the outcome, mispricing, and control-data requirements for the direct-effect regression. Parentheses distinguish US and non-US funds. Panel C reports the AP samples used in the network analyses. Exposed APs have positive global-equity activity in 2023.

Panel A: ETF samples

	US domestic	Non-US, US benchmark	US global	Non-US, non-US benchmark	Total
Full ETF dataset	711	279	575	555	2,120
Cross-market matched universe	55	93	46	100	294
Table 1, Panel B	55	93	33	70	251

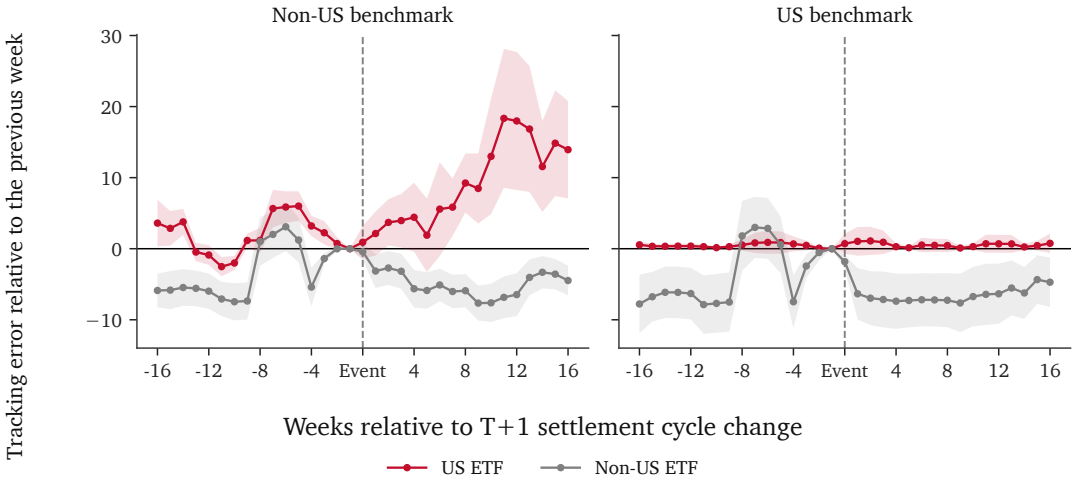
Panel B: Direct-effect pair-day sample

Restriction	Funds (US/non-US)	Pair IDs	Bench.	Days	Fund-days	Pair-days
Synchronized pair file	411 (146/265)	361	120	258	96,635	167,246
Benchmark AUM ≥ $0.1 billion	393 (138/255)	351	112	258	91,732	161,714
Non-US benchmarks	135 (42/93)	97	41	258	32,158	45,468
Baseline complete cases	104 (37/67)	88	37	256	23,069	34,715
Estimation sample	104 (37/67)	88	37	250	23,063	34,709

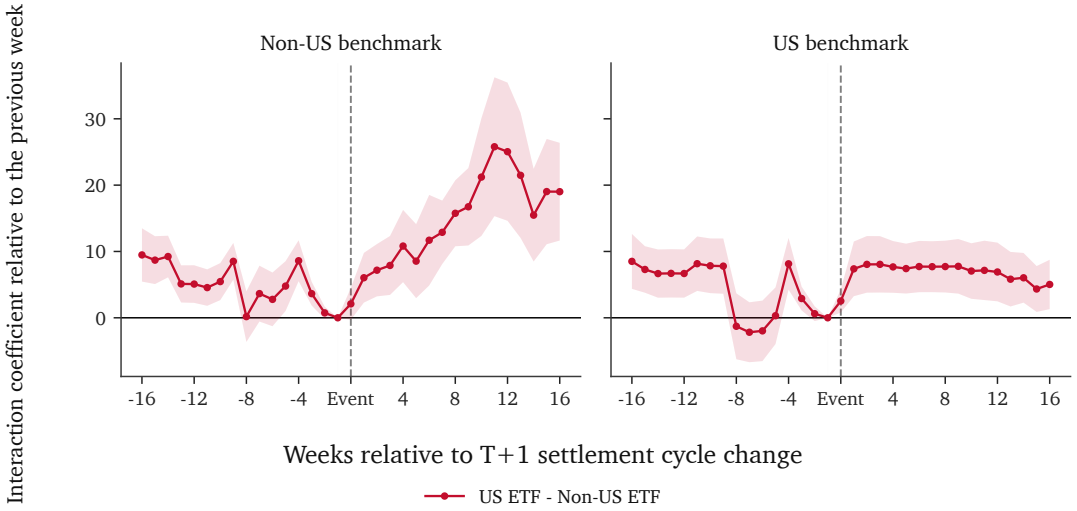
Panel C: Authorized-participant samples

Empirical object	Activity requirement	Exposed	Unexposed	Total APs
AP landscape, 2018–2024	Positive US-listed ETF activity	–	–	39
2023 equity network and composition	Positive equity ETF activity in 2023	20	10	30
2023–2024 gross-activity change	Matched across annual filings	20	28	48
AP-composition RCR row	Non-missing RCR	18	10	28

C Additional figures and tables



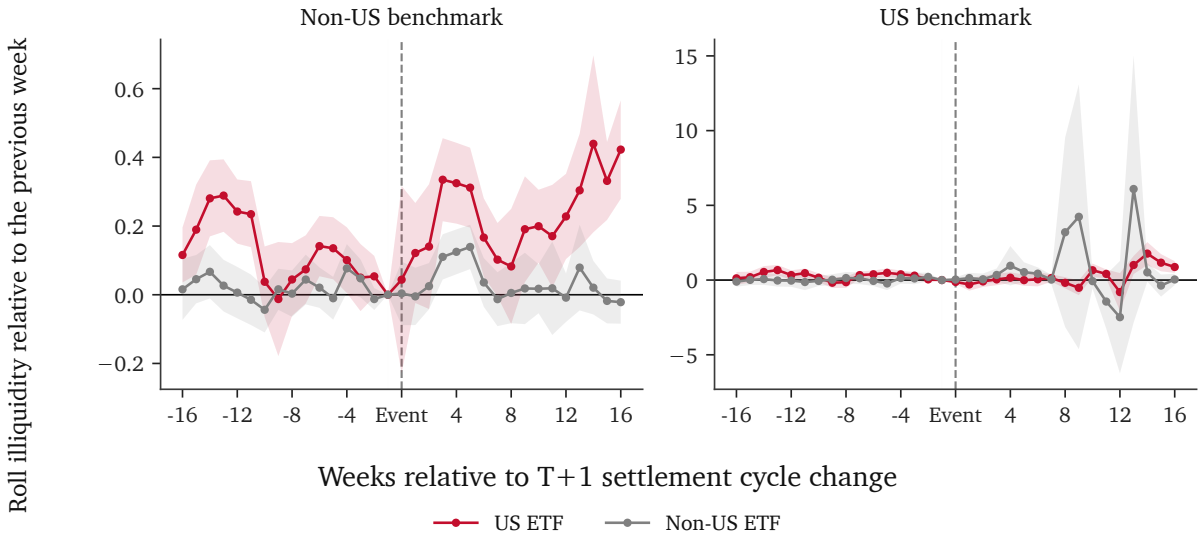
(a) Tracking-error-volatility dynamics



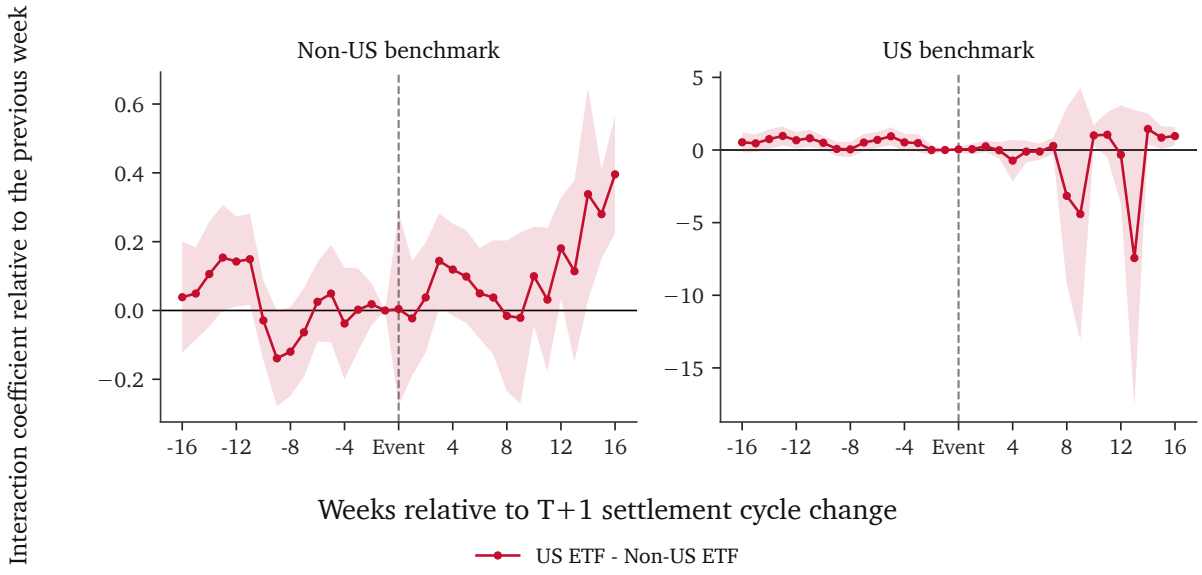
(b) Difference in tracking-error-volatility dynamics

Figure C1. Weekly Tracking-Error Volatility around the T+1 Settlement Cycle Change

This figure plots weekly tracking-error-volatility estimates over the sixteen weeks before and after May 28, 2024. Tracking-error volatility is the standard deviation of fund-minus-benchmark returns over the current and preceding 22 calendar days, requiring at least 10 observations and averaged within each week. Non-US benchmarks appear on the left and US benchmarks on the right. Panel A shows US and non-US ETFs separately; Panel B shows their difference. Estimates include matched-pair fixed effects and are relative to the week before the transition. The dashed line marks the transition, and shaded areas show 95% confidence intervals using standard errors clustered by fund.



(a) Liquidity dynamics



(b) Difference in liquidity dynamics

Figure C2. Weekly ETF Liquidity around the T+1 Settlement Cycle Change

This figure plots weekly ETF illiquidity, measured by Roll, around May 28, 2024. The event window, panel layout, and omitted week follow Figure C1. Panel A shows US and non-US ETFs separately; Panel B shows their difference. Regressions include matched-pair fixed effects. Shaded areas show 95% confidence intervals using standard errors clustered by fund.

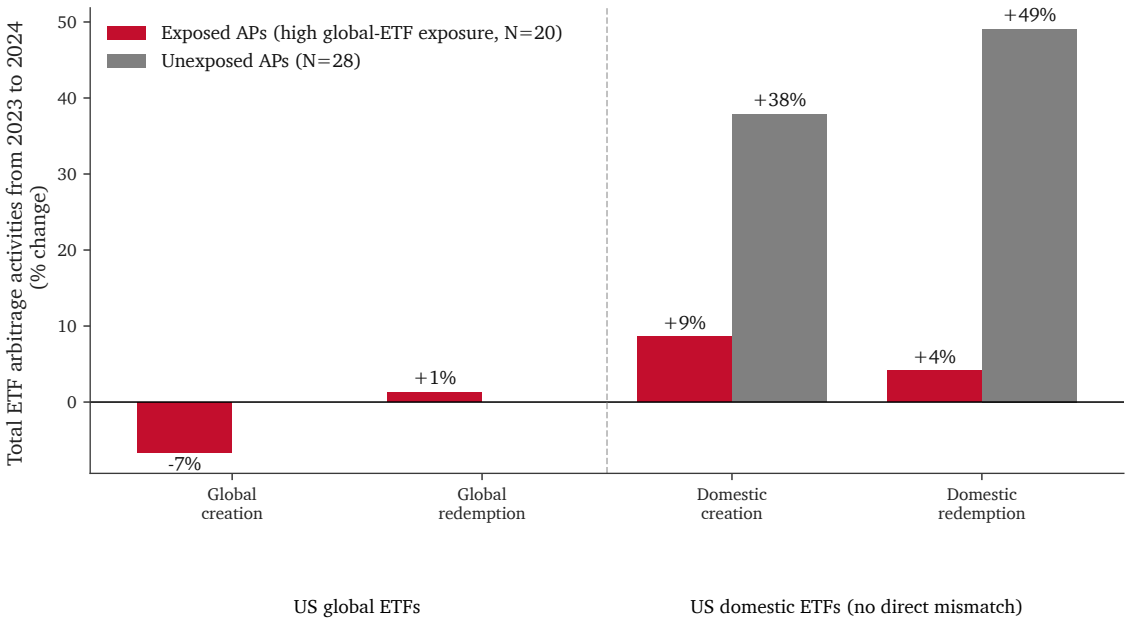


Figure C3. Authorized-Participant Gross Arbitrage Activity Around the T+1 Transition

This figure reports changes in aggregate AP creation and redemption activity from 2023 to 2024, using SEC Form N-CEN. Red bars represent APs with positive global-equity activity in 2023; gray bars represent other APs. The left pair shows global equity ETFs, and the right pair shows domestic equity ETFs. Labels report percentage changes in group totals.

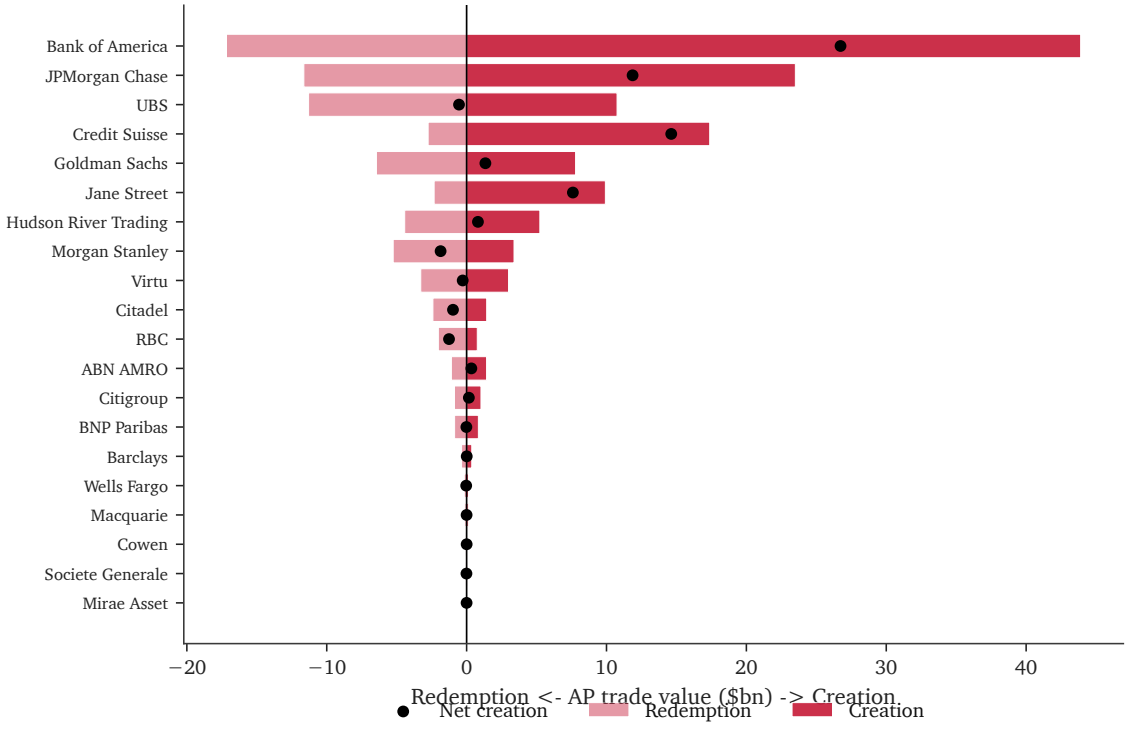


Figure C4. Authorized Participant Creation-Redemption Imbalance in Global Equity ETFs

This figure shows AP creation and redemption values in global equity ETFs in 2023, using SEC Form N-CEN. Redemptions extend left and creations right, in billions of US dollars. Black markers show net creations. APs are ranked by gross creation and redemption value.

Table C1. Settlement Cycles Across Major Jurisdictions

This table reports settlement cycles and their effective dates for major equity markets by market capitalization as of December 31, 2024.

Rank	Jurisdiction	Market Cap (USD Trillions)	Settlement Cycle	Effective Date of Current Rule
1	United States	61.743	T+1	28-May-24
2	European Union	16.904	T+2	October 6, 2014 (CSDR implementation)
3	China	7.68	Mixed: A-shares: T+1, B-shares: T+3	April 15, 2016 (A shares to T+1)
4	Japan	5.215	T+2	16-Jul-19
5	India	3.931	Mixed: T+1, T+0	January 27, 2023 (full transition)
6	United Kingdom	3.686	T+2	6-Oct-14
7	Canada	2.967	T+1	27-May-24
8	Saudi Arabia	2.713	T+2	23-Apr-17
9	Switzerland	2.488	T+2	Long-standing (pre-2014)
10	Taiwan	1.914	T+2	Long-standing (pre-2017)
11	Australia	1.676	T+2	March 2016 (ASX transition)
12	South Korea	1.082	T+2	Long-standing (pre-2017)
13	United Arab Emirates	1.028	T+2	Long-standing (pre-2017)
14	Hong Kong	0.929	T+2	Long-standing (pre-2017)
15	Brazil	0.716	T+2	26-Aug-13
16	Singapore	0.558	T+2	9-Dec-13
17	Mexico	0.482	T+1	28-May-24
18	Russia	0.424	T+2	2-Sep-13
19	Israel	0.388	T+2	Long-standing (pre-2017)
20	Indonesia	0.357	T+2	26-Nov-18

Table C2. Split-Sample Estimates by Authorized Participant Balance-Sheet Capacity

This table estimates arbitrage responsiveness by AP balance-sheet capacity. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest is US × Post × Mispricing. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Columns (1)–(3) cover ETFs served primarily by weak APs; columns (4)–(6) cover those served primarily by strong APs. AP classification follows Table 5; mispricing measures follow Table 3. Regressions include all lower-order interactions and matched-pair and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)					
	Weak APs			Strong APs		
	Mispricing	Mispricing$^+$	Mispricing$^-$	Mispricing	Mispricing$^+$	Mispricing$^-$
	(1)	(2)	(3)	(4)	(5)	(6)
1(US) × **1**(Post) × Mispricing	-0.061***	-0.122***	-0.018	-0.030	0.028	0.004
	(0.019)	(0.020)	(0.027)	(0.027)	(0.039)	(0.036)
1(Post) × Mispricing	0.028*	0.070***	-0.014	0.006	-0.042	-0.004
	(0.016)	(0.018)	(0.020)	(0.023)	(0.026)	(0.028)
1(US) × Mispricing	0.051***	0.075***	0.049**	0.017	-0.023	-0.025
	(0.014)	(0.016)	(0.023)	(0.020)	(0.025)	(0.032)
1(US) × **1**(Post)	0.002	0.001	-0.001	-0.034	-0.003	-0.048
	(0.023)	(0.025)	(0.022)	(0.028)	(0.028)	(0.031)
1(US)	-0.046***	-0.062***	-0.027*	-0.039	-0.062**	-0.027
	(0.016)	(0.019)	(0.016)	(0.026)	(0.029)	(0.028)
ln(TNA)	0.010**	0.010**	0.014***	-0.000	-0.002	0.001
	(0.005)	(0.005)	(0.005)	(0.010)	(0.009)	(0.010)
ln(Fund age)	-0.066***	-0.063***	-0.068***	-0.026	-0.023	-0.035
	(0.014)	(0.014)	(0.014)	(0.029)	(0.027)	(0.029)
Fund returns	0.015*	0.013*	0.014*	-0.019*	-0.018*	-0.018*
	(0.009)	(0.008)	(0.008)	(0.011)	(0.011)	(0.011)
Lagged flow	0.045***	0.045***	0.043***	-0.025	-0.024	-0.024
	(0.015)	(0.014)	(0.015)	(0.028)	(0.026)	(0.026)
Mispricing	-0.006	-0.018	0.019	-0.005	0.039*	0.009
	(0.013)	(0.013)	(0.019)	(0.016)	(0.021)	(0.026)
Roll	0.012	0.012	0.008	0.030	0.033	0.030
	(0.024)	(0.023)	(0.024)	(0.030)	(0.028)	(0.031)
Matched-pair FE	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓
N	24,795	25,283	25,283	8,885	9,364	9,364
R^2	0.0683	0.0693	0.0703	0.0687	0.0683	0.0684

Table C3. Arbitrage Responsiveness by AP Foreign-Exchange Activity

This table estimates arbitrage responsiveness by AP foreign-exchange activity. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest is US × Post × Mispricing. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Lagged mispricing is also included. Columns (1)–(3) cover ETFs served primarily by low-FX-volume APs; columns (4)–(6) cover those served primarily by high-FX-volume APs. High-FX-volume APs are the five largest dealers in the Euromoney FX Survey; ETFs are classified by whether these APs handle at least half of their 2023 activity. Mispricing measures follow Table 3. Regressions include all lower-order interactions and benchmark and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)					
	Low-FX-volume APs			High-FX-volume APs		
	Mispricing	Mispricing$^+$	Mispricing$^-$	Mispricing	Mispricing$^+$	Mispricing$^-$
	(1)	(2)	(3)	(4)	(5)	(6)
1(US) × **1**(Post) × Mispricing	-0.052***	-0.083***	-0.027	-0.074***	-0.092***	-0.031
	(0.018)	(0.025)	(0.027)	(0.022)	(0.026)	(0.032)
1(Post) × Mispricing	0.032**	0.047**	0.013	0.043**	0.066***	-0.010
	(0.016)	(0.020)	(0.023)	(0.019)	(0.019)	(0.030)
1(US) × Mispricing	0.043***	0.061***	0.034	0.040**	0.046**	0.032
	(0.014)	(0.020)	(0.024)	(0.016)	(0.019)	(0.028)
1(US) × **1**(Post)	-0.052**	-0.036	-0.038	0.042*	0.030	0.032
	(0.026)	(0.030)	(0.025)	(0.025)	(0.025)	(0.027)
1(US)	-0.047*	-0.070**	-0.040*	-0.064***	-0.066***	-0.051***
	(0.025)	(0.029)	(0.022)	(0.015)	(0.016)	(0.018)
ln(TNA)	0.019***	0.020***	0.020***	0.009	0.009	0.006
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
ln(Fund age)	-0.050**	-0.053***	-0.045**	-0.077***	-0.074***	-0.073***
	(0.020)	(0.020)	(0.020)	(0.016)	(0.016)	(0.017)
Fund returns	-0.007	-0.007	-0.007	0.015	0.011	0.012
	(0.009)	(0.008)	(0.009)	(0.010)	(0.009)	(0.009)
Lagged flow	0.054***	0.054***	0.053***	0.015	0.014	0.013
	(0.017)	(0.017)	(0.018)	(0.016)	(0.015)	(0.016)
Mispricing	-0.020	-0.019	-0.000	-0.009	-0.017	0.023
	(0.012)	(0.014)	(0.022)	(0.015)	(0.014)	(0.028)
Roll	0.024	0.029	0.019	0.009	0.002	0.006
	(0.022)	(0.022)	(0.021)	(0.030)	(0.030)	(0.031)
Benchmark FE	✓	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓	✓
N	18,697	18,943	18,943	16,325	17,046	17,046
R^2	0.0760	0.0764	0.0765	0.0687	0.0688	0.0680

Table C4. Settlement Cycle Mismatch and Weekly Tracking-Error Volatility

This table estimates changes in weekly tracking-error volatility for US-listed global ETFs relative to non-US ETFs tracking the same benchmark. The dependent variable is the within-week standard deviation of return differences. The variable of interest is US × Post. Controls include weekly averages of one-day lagged log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Columns compare price with benchmark returns, price with NAV returns, and NAV with benchmark returns. Regressions include matched-pair and week fixed effects. Standard errors are clustered by fund and week and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Weekly tracking-error volatility (%)		
	Price versus Benchmark	Price versus NAV	NAV versus Benchmark
	(1)	(2)	(3)
$\mathbf{1}(\text{US}) \times \mathbf{1}(\text{Post})$	0.055***	0.056***	0.051***
	(0.021)	(0.020)	(0.011)
$\mathbf{1}(\text{US})$	-0.118***	-0.249***	0.148***
	(0.017)	(0.017)	(0.009)
ln(TNA)	0.048***	0.033***	0.013***
	(0.006)	(0.006)	(0.004)
ln(Fund age)	-0.095***	-0.060***	-0.023***
	(0.017)	(0.018)	(0.007)
Fund returns	0.018	0.004	0.034***
	(0.020)	(0.020)	(0.010)
Lagged flow	0.013	0.008	0.002
	(0.016)	(0.016)	(0.008)
Roll	-0.044**	-0.079***	0.048***
	(0.018)	(0.018)	(0.011)
Matched-pair FE	✓	✓	✓
Week FE	✓	✓	✓
N	5,314	5,314	5,314
R^2	0.5031	0.4953	0.5699

Table C5. Pre-Transition Composition of Exposed and Unexposed Authorized Participants

This table compares APs with and without positive global-equity activity in 2023. Columns (1) and (2) report means, with standard deviations in parentheses. The final column reports exposed-minus-unexposed differences, with unequal-variance test standard errors in parentheses. Gross activity is creations plus redemptions. Active relationships have positive transactions. Rank and activity shares are averaged across each AP's ETF relationships; HHI measures AP concentration within those ETFs. Netting share measures offsetting creation and redemption activity. RCR is regulatory capital relative to its required minimum. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Exposed APs (1)	Unexposed APs (2)	Difference (1)−(2)
Bank affiliated (%)	65.00 (48.94)	60.00 (51.64)	5.00 (19.66)
Primary dealer at year-end 2023 (%)	55.00 (51.04)	60.00 (51.64)	-5.00 (19.92)
ln(Gross AP activity)	24.447 (2.224)	20.678 (1.933)	3.770*** (0.788)
Number of active ETF relationships	293.15 (271.68)	10.80 (12.19)	282.35*** (60.87)
Mean within-ETF rank percentile (%)	47.55 (18.56)	25.73 (20.79)	21.82** (7.78)
Mean within-ETF activity share (%)	15.46 (10.30)	4.32 (6.74)	11.14*** (3.14)
Mean fund-level AP HHI (%)	28.37 (7.51)	16.99 (6.03)	11.37*** (2.54)
Netting share (%)	69.02 (19.25)	55.30 (32.16)	13.72 (11.04)
ln(RCR)	2.890 (2.426)	3.940 (1.996)	-1.050 (0.852)
Number of APs	20	10	30

Table C6. Spillovers and AP Network Breadth, Concentration, and Composition

This table examines whether AP network characteristics moderate spillovers. The dependent variable is the signed daily percentage change in shares outstanding. The variables of interest interact Indirect Exposure × Post × Mispricing with network characteristics. Exposure and mispricing follow Table 8. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Columns (1)–(4) add network characteristics separately; column (5) includes all four, together with AP capacity, netting, and trading burden. High indicators use the panel medians of 2023 characteristics. Active AP count measures trading relationships; HHI measures concentration in AP trading shares. Non-bank and HFT shares use AP counts. The HFT group comprises Citadel Securities, Hudson River Trading, Jane Street, and Virtu. Regressions include all lower-order interactions and fund and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Existing network characteristics		AP composition		Joint
	Active AP count	AP HHI	Non-bank AP share	HFT AP share	All characteristics
	(1)	(2)	(3)	(4)	(5)
Indirect exposure × Post × Mispricing	-0.047***	0.000	-0.044**	-0.039*	-0.034
	(0.018)	(0.020)	(0.018)	(0.022)	(0.064)
High active AP count × Indirect exposure × Post × Mispricing	0.051				0.031
	(0.045)				(0.073)
High AP HHI × Indirect exposure × Post × Mispricing		-0.060**			-0.007
		(0.024)			(0.063)
High non-bank AP share × Indirect exposure × Post × Mispricing			0.023		-0.005
			(0.037)		(0.053)
High HFT AP share × Indirect exposure × Post × Mispricing				-0.004	0.018
				(0.025)	(0.051)
AP Tier 1 interaction block					✓
Netting and burden interaction blocks					✓
Controls and lower-order interactions	✓	✓	✓	✓	✓
Fund and day FE	✓	✓	✓	✓	✓
ETFs	247	247	247	247	247
N	53,965	53,965	53,965	53,965	53,965
R^2	0.0605	0.0602	0.0598	0.0601	0.0618

Table C7. Alternative Spillover Heterogeneity Specifications

This table examines alternative spillover specifications using combined domestic- and global-equity netting. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest is Indirect Exposure × Post × Mispricing and its interactions with AP characteristics. Exposure and mispricing follow Table 8. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Columns (2)–(4) add AP capacity, concentration, and network channels, respectively; column (5) includes them jointly. Regressions include all lower-order interactions and fund and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)				
	Baseline	AP capacity	AP concentration	AP network	All channels
	(1)	(2)	(3)	(4)	(5)
Indirect exposure × **1**(Post) × Mispricing	-0.041***	-0.082***	0.006	-0.047*	-0.114
	(0.013)	(0.026)	(0.019)	(0.028)	(0.073)
AP capacity × **1**(Post) × Mispricing		-0.096*			-0.101*
		(0.051)			(0.056)
AP capacity × Indirect exposure × Mispricing		-0.089**			-0.071*
		(0.039)			(0.040)
AP capacity × Indirect exposure × **1**(Post) × Mispricing		0.081**			0.071*
		(0.036)			(0.039)
AP concentration × **1**(Post) × Mispricing			0.016		0.002
			(0.042)		(0.054)
AP concentration × Indirect exposure × Mispricing			0.077***		0.039
			(0.027)		(0.042)
AP concentration × Indirect exposure × **1**(Post) × Mispricing			-0.057***		0.018
			(0.020)		(0.035)
AP netting friction × **1**(Post) × Mispricing				-0.009	-0.003
				(0.061)	(0.069)
AP netting friction × Indirect exposure × Mispricing				-0.064*	-0.055
				(0.034)	(0.056)
AP netting friction × Indirect exposure × **1**(Post) × Mispricing				0.073*	0.101*
				(0.040)	(0.055)
AP global exposure × **1**(Post) × Mispricing				0.015	0.026
				(0.066)	(0.059)
AP global exposure × Indirect exposure × Mispricing				-0.078*	-0.079**
				(0.041)	(0.033)
AP global exposure × Indirect exposure × **1**(Post) × Mispricing				0.056	0.053
				(0.044)	(0.037)
AP netting friction × AP global exposure × Indirect exposure × Mispricing				0.168***	0.141*
				(0.063)	(0.073)
AP netting friction × AP global exposure × Indirect exposure × **1**(Post) × Mispricing				-0.198***	-0.176***
				(0.058)	(0.061)
Controls	✓	✓	✓	✓	✓
Lower-order interactions	✓	✓	✓	✓	✓
Fund FE	✓	✓	✓	✓	✓
Day FE	✓	✓	✓	✓	✓
N	59,541	53,965	59,541	59,541	53,965
R^2	0.0590	0.0608	0.0595	0.0595	0.0613

Table C8. Event-Time Dynamics of Primary Market Arbitrage Responsiveness

This table estimates event-time changes in arbitrage responsiveness using equation (2). The dependent variable is the signed daily percentage change in shares outstanding. The variables of interest interact US and Mispricing with event-period indicators, replacing Post. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Event periods span 21 trading days over a $[-90, +90]$ window around May 28, 2024; the omitted period is $[-21, -1]$. Columns use the mispricing measures in Table 3. Joint tests report p-values for pre- and post-transition interactions separately. Regressions include all lower-order interactions and matched-pair and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)		
	Mispricing	Mispricing^{+}	Mispricing^{-}
	(1)	(2)	(3)
Pre-transition			
$[-90, -85]$	0.031	0.013	-0.041
	(0.048)	(0.041)	(0.067)
$[-84, -64]$	-0.049	-0.007	-0.068*
	(0.041)	(0.036)	(0.040)
$[-63, -43]$	-0.015	0.009	-0.028
	(0.041)	(0.039)	(0.062)
$[-42, -22]$	0.009	0.016	-0.004
	(0.047)	(0.052)	(0.053)
$[-21, -1]$	—	—	—
Post-transition			
$[0, +20]$	-0.061	-0.082**	-0.057
	(0.041)	(0.035)	(0.054)
$[+21, +41]$	-0.032	-0.173***	-0.022
	(0.048)	(0.062)	(0.053)
$[+42, +62]$	-0.017	-0.062	0.051
	(0.047)	(0.060)	(0.050)
$[+63, +83]$	-0.018	-0.042	-0.015
	(0.040)	(0.040)	(0.042)
$[+84, +90]$	-0.113***	-0.110**	-0.182**
	(0.037)	(0.049)	(0.075)
Joint test, pre-transition (p)	0.206	0.958	0.551
Joint test, post-transition (p)	0.002	< 0.001	0.085
Pair FE	Yes	Yes	Yes
Day FE	Yes	Yes	Yes
Lagged controls	Yes	Yes	Yes
N	24,288	24,977	24,977
R^2	0.0601	0.0604	0.0648

Table C9. Placebo Transition Dates

This table estimates equation (2) using placebo transition dates and only pre-transition observations. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest is US × placebo-Post × Mispricing. Controls include one-day lags of log total net assets, log fund age, fund returns, the percentage change in shares outstanding, and Roll illiquidity, which are defined in Table 3. Placebo dates are 80, 60, 40, and 20 trading days before May 28, 2024. Specifications use signed mispricing and five-day maximum premiums. Regressions include all lower-order interactions and matched-pair and day fixed effects. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Placebo event date (trading days before 28 May 2024)			
	−80	−60	−40	−20
	(1)	(2)	(3)	(4)
Mispricing	0.002	0.028	0.045	0.008
	(0.029)	(0.028)	(0.031)	(0.037)
Mispricing^{+}	0.009	0.002	0.004	-0.008
	(0.032)	(0.033)	(0.039)	(0.026)
Pair FE	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes
Lagged controls	Yes	Yes	Yes	Yes
N	17,760	17,760	17,760	17,760

Table C10. Historical Audit: The 2017 T+3-to-T+2 Transition

This table estimates equation (2) around the September 5, 2017 US transition from T+3 to T+2. The dependent variable is the signed daily percentage change in shares outstanding. The variable of interest is US × Post × Mispricing. Controls include one-day lags of log total net assets and log fund age, which are defined in Table 3. Synchronized mispricing uses TAQ prices and Morningstar NAVs and is standardized within the sample. Matches use the 2024 benchmark mapping, observed within one year before and after the transition. Column (1) includes matched-pair and day fixed effects; column (2) replaces pair with fund fixed effects; column (3) restricts matches to European venues; column (4) omits controls. Regressions include all lower-order interactions. Standard errors are clustered by fund and day and are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

	Dependent variable: Change in shares outstanding (%)			
	(1)	(2)	(3)	(4)
1(US) × **1**(Post) × Mispricing	0.048	0.039	0.048	0.036
	(0.076)	(0.082)	(0.075)	(0.076)
1(Post) × Mispricing	-0.051	-0.053	-0.051	-0.048
	(0.070)	(0.072)	(0.069)	(0.071)
1(US) × Mispricing	-0.084**	-0.074*	-0.084**	-0.072*
	(0.039)	(0.042)	(0.039)	(0.039)
1(US) × **1**(Post)	-0.057	-0.071*	-0.057	-0.055
	(0.041)	(0.039)	(0.041)	(0.041)
1(US)	0.035		0.035	-0.009
	(0.038)		(0.036)	(0.033)
Mispricing	0.094***	0.091***	0.094***	0.091***
	(0.032)	(0.034)	(0.032)	(0.032)
ln(TNA)	-0.028**	-0.306***	-0.028**	
	(0.012)	(0.078)	(0.011)	
ln(Fund age)	0.032	0.194	0.032	
	(0.051)	(0.283)	(0.050)	
Pair FE	Yes	No	Yes	Yes
Fund FE	No	Yes	No	No
Day FE	Yes	Yes	Yes	Yes
Lagged controls	Yes	Yes	Yes	No
Clean synchronization only	No	No	Yes	No
N	36,951	36,951	36,951	36,951
R^2	0.0269	0.0297	0.0269	0.0266

D Constructing Time-Synchronized Mispricing

This appendix describes how I align mispricing measures for US-listed global ETFs and their matched non-US counterparts.

The timing problem. Conventional end-of-day mispricing is $\mathrm{dp}^{\mathrm{EOD}}_{f,t} = P^{\mathrm{US}}_{f}(16{:}00\ \mathrm{ET})/\mathrm{NAV}_{f,t} - 1$. For ETFs holding foreign securities, the US closing price can incorporate information arriving after the underlying markets close. European markets, for example, close around 11:30 ET. The end-of-day deviation can therefore combine mispricing with differences in valuation timing.

Construction. Each matched pair consists of a US-listed global ETF and a non-US ETF tracking the same benchmark, both denominated in US dollars. Because the data do not report the US ETF's intraday NAV at the foreign-market close, I use its counterpart's local-close NAV as a same-benchmark valuation proxy. I sample the US price at that close when trading hours overlap and use the closest available US price otherwise. For each pair (f, f'), synchronized mispricing is

$$\mathrm{dp}^{\mathrm{sync2}}_{f,t} \;=\; \frac{P^{\mathrm{US}}_{f}\left(\tau_{f'}\right)}{\kappa_{f,f'}\,\mathrm{NAV}_{f',t}} \;-\; 1,$$

where $\tau_{f'}$ is the US-price sampling time associated with the non-US counterpart's local close, and $\kappa_{f,f'}$ adjusts for differences in share-class scale. For non-US control observations, I use the fund's own local-close price and NAV.

Scaling. I estimate $\kappa_{f,f'}$ as the median raw NAV ratio within each pair, dividing the series into separate segments at share splits. Because both funds are denominated in US dollars, the factor adjusts share-class levels. The typical pair's daily NAV ratio lies within 37 basis points of its scaling factor.

Figure D1 illustrates the construction for a European counterpart. Panel A compares the US closing price with an earlier NAV observation. Panel B aligns the US price with the counterpart's closing NAV, around 11:30 ET. Green markers identify the synchronized observations; red shading marks the timing gap.

Robustness checks. Estimating the scaling factor using pre-transition observations only, or restricting the sample to pairs without share-split segmentation, produces a more negative coefficient. The estimate also remains negative when I use conventional end-of-day mispricing for both groups or restrict the sample to European matches with overlapping trading hours. Section 4.2 reports the alternative estimates.

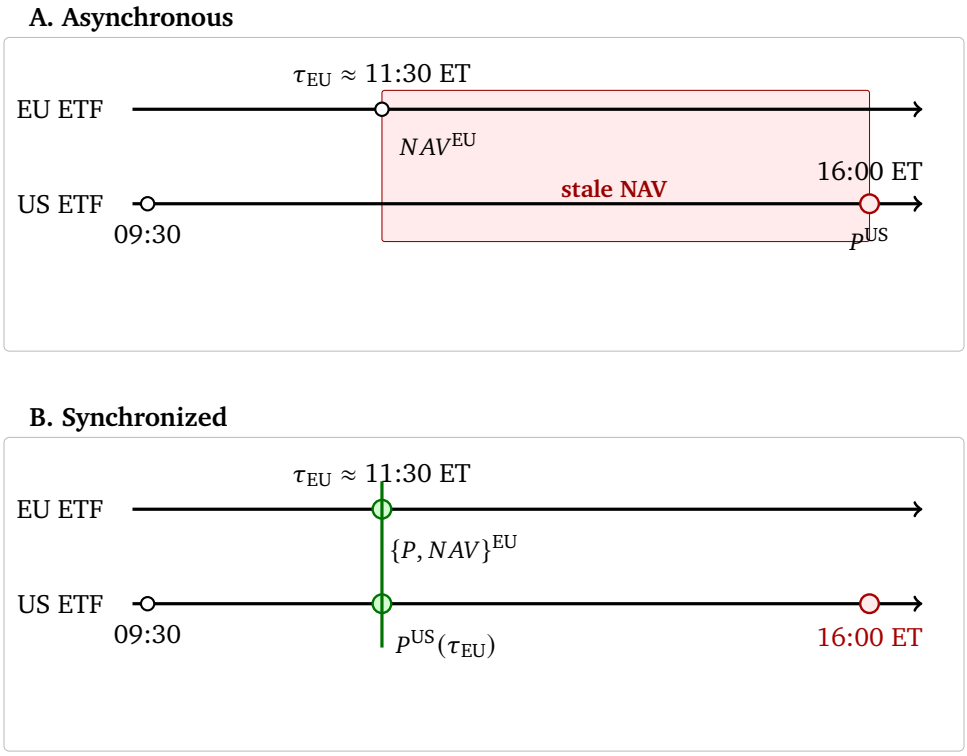


Figure D1. Timing of price–NAV observations for conventional and synchronized measures.

Panel A pairs the US closing price with an earlier NAV observation. Panel B aligns the US price with the European ETF's closing NAV. Green markers identify the synchronized observations; red shading marks the timing gap.

E Model Derivations and Extensions

This appendix derives the responsiveness expression (19), proves Propositions 1 and 2, and develops the extensions omitted from the main text. Throughout the one-AP derivation, AP a is active in both pairs, so the first-order conditions (17) hold with $x_{aX}, x_{aD} > 0$. I suppress the a subscript on φ, ω, Γ, and Ψ where no ambiguity arises, and write $\Psi'' \equiv \omega > 0$ and Ψ''' for the derivatives of the cost function evaluated at Γ_a.

E.1 Individual AP Responsiveness and One-AP Proofs

Step 1: the slope of arbitrage supply. Write the two first-order conditions (17) as $m_f = 2b_f x_{af} + 2\bar{p}_f C_{af} + \varphi_{af}\Psi'(\Gamma_a)$ for $f \in \{X, D\}$, with Γ_a given by (8). Totally differentiating with respect to (m_X, m_D) and using $\partial\Gamma_a/\partial x_{af} = \varphi_{af}$ gives

$$\begin{pmatrix} 2b_X + \varphi_{aX}^2\omega & \varphi_{aX}\varphi_{aD}\omega \\ \varphi_{aX}\varphi_{aD}\omega & 2b_D + \varphi_{aD}^2\omega \end{pmatrix}\begin{pmatrix} dx_{aX} \\ dx_{aD} \end{pmatrix} = \begin{pmatrix} dm_X \\ dm_D \end{pmatrix}.$$

Call this matrix $\mathbf{J}_a$; it is the local supply Jacobian obtained from the first-order conditions, and its off-diagonal element is the cost complementarity (10). Its determinant is

$$\det \mathbf{J}_a = \left(2b_X + \varphi_{aX}^2\omega\right)\left(2b_D + \varphi_{aD}^2\omega\right) - \varphi_{aX}^2\varphi_{aD}^2\omega^2 = 4b_X b_D + 2\omega\left(b_D\varphi_{aX}^2 + b_X\varphi_{aD}^2\right), \tag{A1}$$

which is strictly positive for any $\omega > 0$ given $b_X, b_D > 0$. For completeness, the Hessian of the negated objective is

$$\begin{pmatrix} 4b_X + \varphi_{aX}^2\omega & \varphi_{aX}\varphi_{aD}\omega \\ \varphi_{aX}\varphi_{aD}\omega & 4b_D + \varphi_{aD}^2\omega \end{pmatrix},$$

whose determinant is $16b_X b_D + 4\omega(b_D\varphi_{aX}^2 + b_X\varphi_{aD}^2) > 0$. Hence each AP's objective is strictly concave in its own two choices and its interior best response is unique. The factor-of-two difference between the $2b_f$ term in the supply Jacobian and the $4b_f$ term in the negative objective Hessian is intentional. The supply Jacobian differentiates the markup relation as realized m_f varies across demand shocks. The objective Hessian instead varies x_{af} while holding the demand imbalance fixed, so it also differentiates

$m_f(x_{af})$. Inverting $\mathbf{J}_a$ gives

$$\lambda_{aD} \equiv \frac{\partial x_{aD}}{\partial m_D} = \frac{2b_X + \varphi_{aX}^2 \omega}{\det \mathbf{J}_a} = \frac{1}{2b_D + \varphi_{aD}^2\, \omega\, \kappa_{aD}}, \qquad \kappa_{aD} = \frac{2b_X}{2b_X + \varphi_{aX}^2 \omega} \in (0,1), \tag{A2}$$

where the second equality follows by dividing numerator and denominator of the first by $2b_X + \varphi_{aX}^2\omega$ and using (A1). This is (19); the expression for λ_{aX} is symmetric. Because $\kappa_{aD} < 1$, the AP's cross-pair adjustment damps the effect of shared cost on the domestic slope without eliminating it, and λ_{aD} is strictly decreasing in ω and in φ_{aD}.

Step 2: the direct effect (Proposition 1). By the symmetric counterpart of (A2), $\lambda_{aX} = \left(2b_X + \varphi_{aX}^2 \omega\, \kappa_{aX}\right)^{-1}$ with $\kappa_{aX} = 2b_D/(2b_D + \varphi_{aD}^2\omega)$. A wider bridge raises φ_{aX} by (13), with $\partial\varphi_{aX}/\partial h_t = \gamma_\varphi > 0$. Differentiating the denominator gives

$$\frac{d\lambda_{aX}}{dh_t} = -\lambda_{aX}^2 \left[2\gamma_\varphi \varphi_{aX}\, \omega\, \kappa_{aX} + \varphi_{aX}^2\, \kappa_{aX}^2\, \Psi''' \frac{d\Gamma_a}{dh_t} \right]. \tag{A3}$$

The first term is strictly positive whenever $\omega > 0$, so convexity signs the mechanical partial effect holding usage fixed. The total derivative is negative if and only if the bracket is positive. That condition holds for a quadratic cost (where $\Psi''' = 0$) and under the negative-spillover regime in Step 3, but convexity alone does not rule out a sufficiently negative curvature term. This proves the conditional statement in (20). Under the power-cost parameterization, the T+2-weight and capacity comparative statics are therefore read holding the other primitives and the relevant usage response fixed; they are not consequences of convexity alone.

Step 3: the spillover (Proposition 2). The settlement bridge affects domestic responsiveness through the shared cost and the cross-border capacity intensity. Thus φ_{aD} and b_D are unaffected, while h_t enters (A2) through ω and φ_{aX}. Write $D_{aD} \equiv 2b_D + \varphi_{aD}^2 \omega \kappa_{aD}$, so $\lambda_{aD} = 1/D_{aD}$ and $d\lambda_{aD}/dh_t = -\lambda_{aD}^2\, dD_{aD}/dh_t$. Substituting $\kappa_{aD} = 2b_X/(2b_X + \varphi_{aX}^2\omega)$,

$$\varphi_{aD}^2\, \omega\, \kappa_{aD} = \frac{2b_X \varphi_{aD}^2\, \omega}{2b_X + \varphi_{aX}^2 \omega},$$

and differentiating this expression with respect to ω and φ_{aX} separately gives

$$\frac{\partial}{\partial\omega}\left(\varphi_{aD}^2\omega\kappa_{aD}\right) = \varphi_{aD}^2\,\kappa_{aD}^2, \qquad \frac{\partial}{\partial\varphi_{aX}}\left(\varphi_{aD}^2\omega\kappa_{aD}\right) = -\frac{\varphi_{aD}^2\,\varphi_{aX}\,\omega^2\,\kappa_{aD}^2}{b_X}.$$

Since $d\omega/dh_t = \Psi'''\, d\Gamma_a/dh_t$ and $d\varphi_{aX}/dh_t = \gamma_\varphi$, the chain rule yields

$$\frac{d\lambda_{aD}}{dh_t} = -\lambda_{aD}^2\,\varphi_{aD}^2\,\kappa_{aD}^2\left[\Psi'''\,\frac{d\Gamma_a}{dh_t} - \frac{\gamma_\varphi\,\varphi_{aX}\,\omega^2}{b_X}\right], \tag{A4}$$

which is (21). Because $\lambda_{aD}^2\varphi_{aD}^2\kappa_{aD}^2 > 0$, the spillover is strictly negative if and only if the bracket is strictly positive. Under the maintained condition $\Psi''' > 0$, a negative spillover requires $d\Gamma_a/dh_t > 0$, but that increase in aggregate usage is not sufficient: the second term is strictly positive whenever $\gamma_\varphi > 0$, so a sufficiently large increase in cross-border capacity intensity—which induces the AP to shed cross-border positions and frees balance sheet for the domestic book—can outweigh the curvature channel. At $\Psi''' = 0$ the bracket is strictly negative and the predicted spillover is positive, which is the reallocation prediction recovered exactly when the mechanism of this paper is switched off. Under the power cost in (A18) with $\eta > 2$, we have $\Psi''' > 0$, and the sign of the bracket is then an empirical matter; the estimates in Section 4.5 are what identify the regime. □

Step 4: the sign of $d\Gamma_a/dh_t$. The necessary condition $d\Gamma_a/dh_t > 0$ is not automatic, because the reform both adds predetermined usage and induces the AP to release discretionary usage. Totally differentiating (8) gives

$$\frac{d\Gamma_a}{dh_t} = \underbrace{\gamma_\Lambda R_a}_{\text{predetermined load}} + \underbrace{\gamma_\varphi x_{aX}}_{\text{intensity}} + \varphi_{aX}\frac{dx_{aX}}{dh_t} + \varphi_{aD}\frac{dx_{aD}}{dh_t},$$

using (15) and (13). The first two terms are strictly positive, while the last two equilibrium adjustments need not have known signs without the conditions above. Hence the exact condition is

$$\gamma_\Lambda R_a + \gamma_\varphi x_{aX} > -\varphi_{aX}\frac{dx_{aX}}{dh_t} - \varphi_{aD}\frac{dx_{aD}}{dh_t},$$

which is (22). It holds when the predetermined-load and intensity increments dominate the total discretionary release. Because $R_a = F_a(1 - n_a)$, Equation (15) makes the predetermined increment

larger for APs with greater gross pipeline scale F_a and poorer netting n_a. The cross-partial (16) follows immediately from its multiplicative form.

E.2 AP Replacement and Participation

Let $\mathcal{X}_D = \{a \in \mathcal{A}_D : z_a > 0\}$ and $\mathcal{U}_D = \{a \in \mathcal{A}_D : z_a = 0\}$ denote APs with positive and zero structural exposure. The exact zero is a benchmark; empirically, potential substitutes can have lower but positive exposure. To distinguish a change in an AP's conditional slope from movement along that slope, hold the daily active set fixed and evaluate all slopes at a common pair of signals (m_D, m_X).

Proposition A1 (Quantity replacement and the conditional slope)**.** *Suppose the settlement bridge lowers the domestic slope of each positive-exposure AP and leaves each zero-exposure AP's schedule unchanged:*

$$\frac{\partial \lambda_{aD}}{\partial h_t} < 0 \quad (a \in \mathcal{X}_D), \qquad \left.\frac{\partial \lambda_{jD}}{\partial h_t}\right|_{m_D, m_X} = 0 \quad (j \in \mathcal{U}_D). \tag{A5}$$

Then aggregate primary-market responsiveness falls:

$$\frac{\partial \Lambda_D^P}{\partial h_t} = \sum_{a \in \mathcal{X}_D} \frac{\partial \lambda_{aD}}{\partial h_t} < 0. \tag{A6}$$

An active zero-exposure AP can nevertheless increase the quantity it supplies in response to a domestic demand shock because the equilibrium deviation becomes more responsive to that shock.

The first result follows directly by differentiating (24). For the quantity result, let Θ_D denote the secondary-market slope and define $\Lambda_D^T = \Lambda_D^P + \Theta_D$. The equilibrium response of AP j's position to a domestic demand shock is

$$\frac{\partial x_{jD}}{\partial u_D} = \lambda_{jD} \frac{2b_D}{1 + 2b_D \Lambda_D^T}. \tag{A7}$$

Holding λ_{jD} and Θ_D fixed gives

$$\frac{\partial}{\partial h_t}\left(\frac{\partial x_{jD}}{\partial u_D}\right) = -\frac{(2b_D)^2 \lambda_{jD}}{\left(1 + 2b_D \Lambda_D^T\right)^2} \frac{\partial \Lambda_D^P}{\partial h_t} > 0. \tag{A8}$$

Thus an alternative AP can absorb more quantity without becoming more responsive to a given deviation. Full slope replacement requires some alternative AP schedules to become more elastic.

To allow such responses, let $\mathcal{I}_D$ collect APs whose slopes fall and $\mathcal{G}_D$ collect APs whose slopes rise.

Define

$$L_D = -\sum_{a \in \mathcal{I}_D} \frac{\partial \lambda_{aD}}{\partial h_t} > 0, \qquad R_D = \sum_{j \in \mathcal{G}_D} \frac{\partial \lambda_{jD}}{\partial h_t} \geq 0, \qquad \rho_D = \frac{R_D}{L_D}. \tag{A9}$$

Then

$$\frac{\partial \Lambda_D^P}{\partial h_t} = -(1 - \rho_D) L_D. \tag{A10}$$

The benchmark has $\rho_D = 0$, partial slope replacement has $0 < \rho_D < 1$, and only $\rho_D \geq 1$ eliminates the aggregate slope loss. Annual AP volume can reveal quantity compensation and participation changes, but it does not identify ρ_D.

Over longer horizons, relationships can enter or exit. Let $\mathcal{C}_D$, $\mathcal{E}_D$, and $\mathcal{L}_D$ denote continuing, entering, and leaving AP relationships. The aggregate change decomposes as

$$\Delta \Lambda_D^P = \sum_{a \in \mathcal{C}_D} \Delta \lambda_{aD} + \sum_{a \in \mathcal{E}_D} \lambda_{aD,post} - \sum_{a \in \mathcal{L}_D} \lambda_{aD,pre}. \tag{A11}$$

The first term is the continuing-AP intensive margin. The last two are participation margins.

E.3 Secondary-Market Arbitrage and Equilibrium Mispricing

Let $G_D^S(m_D, h_t; m_X)$ denote secondary-market arbitrage supply and define

$$\Theta_D = \frac{\partial G_D^S}{\partial m_D} \geq 0, \qquad \Lambda_D^T = \Lambda_D^P + \Theta_D. \tag{A12}$$

Domestic market clearing is

$$m_D = 2b_D \left[u_D - G_D^P(m_D, h_t; m_X) - G_D^S(m_D, h_t; m_X) \right]. \tag{A13}$$

The implicit-function theorem gives

$$\frac{\partial m_D^*}{\partial u_D} = \frac{2b_D}{1 + 2b_D \Lambda_D^T}, \qquad \frac{\partial X_D^P}{\partial u_D} = \Lambda_D^P \frac{2b_D}{1 + 2b_D \Lambda_D^T}, \qquad \left. \frac{\partial X_D^P}{\partial m_D^*} \right|_{u\text{-variation}} = \Lambda_D^P. \tag{A14}$$

Secondary-market supply changes how much mispricing a demand shock produces, but it does not enter the primary-market slope measured by changes in shares outstanding.

Holding u_D and m_X fixed, the conditional level effect is

$$\left.\frac{dm_D^*}{dh_t}\right|_{u_D,m_X} = -\frac{2b_D}{1+2b_D\Lambda_D^T}\left[\left.\frac{\partial G_D^P}{\partial h_t}\right|_{m_D,m_X} + \left.\frac{\partial G_D^S}{\partial h_t}\right|_{m_D,m_X}\right]. \tag{A15}$$

Equilibrium mispricing therefore depends on the shift in total arbitrage supply, not on the primary-market slope alone. With nonlinear Ψ_a, no global level–slope restriction follows from the local derivative in (A14).

E.4 Creation and Redemption States

The symmetric core model does not by itself predict a premium–discount asymmetry. Let $q \in \{+,-\}$ index a premium, associated with creation, and a discount, associated with redemption. A directional extension replaces (13) with

$$\varphi_{aX}^{+} = \varphi_{aX}^{+,0} + \gamma_+ h_{at}, \qquad \varphi_{aX}^{-} = \varphi_{aX}^{-,0} + \gamma_- h_{at}\zeta_t, \qquad \gamma_+, \gamma_- > 0, \tag{A16}$$

where $\zeta_t \geq 0$ measures the cost of temporarily funding the redemption gap. Asset sourcing can bind routinely for creations, while redemption intensity rises when the settlement interval spans a weekend or funding pressure is high. The slope formulas apply state by state after replacing φ_{aX} with φ_{aX}^q.

E.5 Nesting the Gorbatikov–Sikorskaya Benchmark

Setting $\Psi_a \equiv 0$ recovers the Gorbatikov and Sikorskaya (2021) primary-market problem. If secondary supply is also zero, aggregate primary-market supply and its slope are

$$X_f^P = \frac{1}{2b_f}\left[N_{f,act}m_f - 2\bar{p}_f\sum_a C_{af}\right], \qquad \Lambda_f^{P,GS} = \frac{N_{f,act}}{2b_f}. \tag{A17}$$

Pair-specific proportional costs affect the intercept. The slope changes only when the active AP set changes. The shared convex cost adds the continuing-AP intensive margin represented by the first term of (A11).

E.6 Alternative Balance-Sheet Cost Functions

The main text imposes the local cost-shape conditions directly. A power specification consistent with Eisenbach and Phelan (2026) is

$$\Psi_a(\Gamma) = \frac{\psi_a}{\eta}\frac{\Gamma^{\eta}}{B_a^{\eta-1}}, \qquad \psi_a > 0, \quad \eta > 2, \tag{A18}$$

where B_a indexes effective capacity. It implies

$$\Psi_a''(\Gamma_a) = \psi_a(\eta-1)\frac{\Gamma_a^{\eta-2}}{B_a^{\eta-1}}, \qquad \frac{\partial \Psi_a''}{\partial B_a} < 0, \qquad \frac{\partial \Psi_a''}{\partial \Gamma_a} > 0. \tag{A19}$$

Greater effective capacity therefore reduces the sensitivity of marginal cost to additional usage, holding other primitives fixed.

A shrinking-buffer specification motivated by Buchak et al. (2024) is

$$\Psi_a^B(\Gamma) = -\psi_a B_a \log\left(1 - \frac{\Gamma}{B_a}\right), \qquad \Gamma < B_a, \tag{A20}$$

for which

$$(\Psi_a^B)'''(\Gamma) = \frac{2\psi_a B_a}{(B_a - \Gamma)^3} > 0. \tag{A21}$$

Neither proposition requires these global functional forms. The results use only the local cost-shape and dominance conditions stated in the main text.